

Ad hoc semantic reconciliation by cognitive agents: a four-setting synthesis

Programme abstract: Two software systems that must exchange information rarely share a model of the world. The classical answer is to agree a common standard in advance; this programme asks whether two systems that can reason can instead reconcile their divergent models for the occasion, ad hoc, machine-to-machine, with no standard settled beforehand. Across four operational settings in network and service management (configuration, intent, cross-domain provisioning, and observability), it builds a measured harness, scores every reconciliation against a validated gold standard, and varies one master control: where the cognition sits, from two live reasoning agents through one inert to both inert. The through-line the four settings establish is a single thesis. It is cognition that completes a reconciliation. Descriptor methods (matching names, then names plus a gloss) carry it to a ceiling and stop; where both systems can reason, the remainder is closed autonomously, with no agreed model and no human in the loop. A thin, published reference can substitute for cognition, or supply an inert side the facts it lacks, but its reach ends there: it hands over information, never the authority to decide, whose value governs or whether a trade-off is acceptable. And the pragmatic layer (what a reconciled thing is for, whether it matters, who decides) is the frontier the descriptor methods never reach, decisive for meaning and itself bounded by the agent's capability. This document is the synthesis that sits atop the four setting reports: it introduces the concepts once, distils each setting to its essentials, and gathers the findings so they can be read as one result. The full evidence for any setting lives in its own report (1/4–4/4), to which this one points throughout.

Part I: The idea

1. The problem: two models, and no time to standardise

When two software systems must work together but hold different models (of the same domain, or of adjacent domains that must connect), someone has to reconcile those models. The scope a model must cover is not fixed, the useful level of abstraction depends on the task, and capable engineers, working independently, produce different but defensible models of the same network. The classical remedy is a universal standard agreed ahead of time: everyone adopts one vocabulary, and the reconciliation is done once, by committee, before the systems ever meet. That remedy is expensive, slow, and permanently behind the systems it tries to govern.

This programme investigates the alternative that becomes available once the systems on each side can **reason**. Two such systems need not wait for a standard; they can reconcile their models **ad hoc**: between themselves, for the task at hand, machine-to-machine. The question is whether that actually works, how far it reaches, and what (if anything) a small published aid adds to it. The answer is built empirically, in a harness where the reconciling parties are language-model agents and every outcome is scored against a validated gold standard, so that the claims rest on measurement across conditions rather than a single worked demonstration.

2. The lift: from a data model to a portable semantic model

The move the whole programme turns on is the **lift**, and it is worth drawing before anything else (Figure 1).

A system usually holds its content as a **data model**: a schema and its records as they sit in a database or

a controller. That content is already a partial semantic picture, but a fixed one, frozen by the format it uses: a field named `och_grade` with values `1`, `2`, `3` is legible to the software and engineers built around its schema, yet says nothing, on its own, to a consumer that was not told what it means; the meaning lives in a specification and in shared convention, outside the data itself. The **lift** is the move from that to a **semantic model**: the same content given grounded, explicit meaning, structured as three parts. First, an **ontology**, which includes its **lexicon**: a schematic layer (the concepts, their kinds and relations, and the preferred labels and synonyms that name them) and a concrete layer (the individual instances that populate those concepts). Second, **pragmatics**: the contextual information a consumer needs (what a thing is for, whose authority governs it, in what context it holds). Third, **provenance**: who asserted each part, by what method, and how firmly. The lift itself is a **cognitive act**, helped along by a few **supports** that are aids to that cognition rather than parts of the model: definitions, worked examples, a canonical example, and an optional link to a shared reference. Who performs the lift is itself a variable of the study: a live system lifts and explains itself; an inert one is lifted by whatever cognition reads it. In the benchmark cases that self-lift is provided as a materialised artefact, so the reconciliation runs over a fixed substrate; §13.8 lifts that assumption (an agent produces the lift from the schema surface alone) and finds the reconciliation unchanged where the study leans on it.

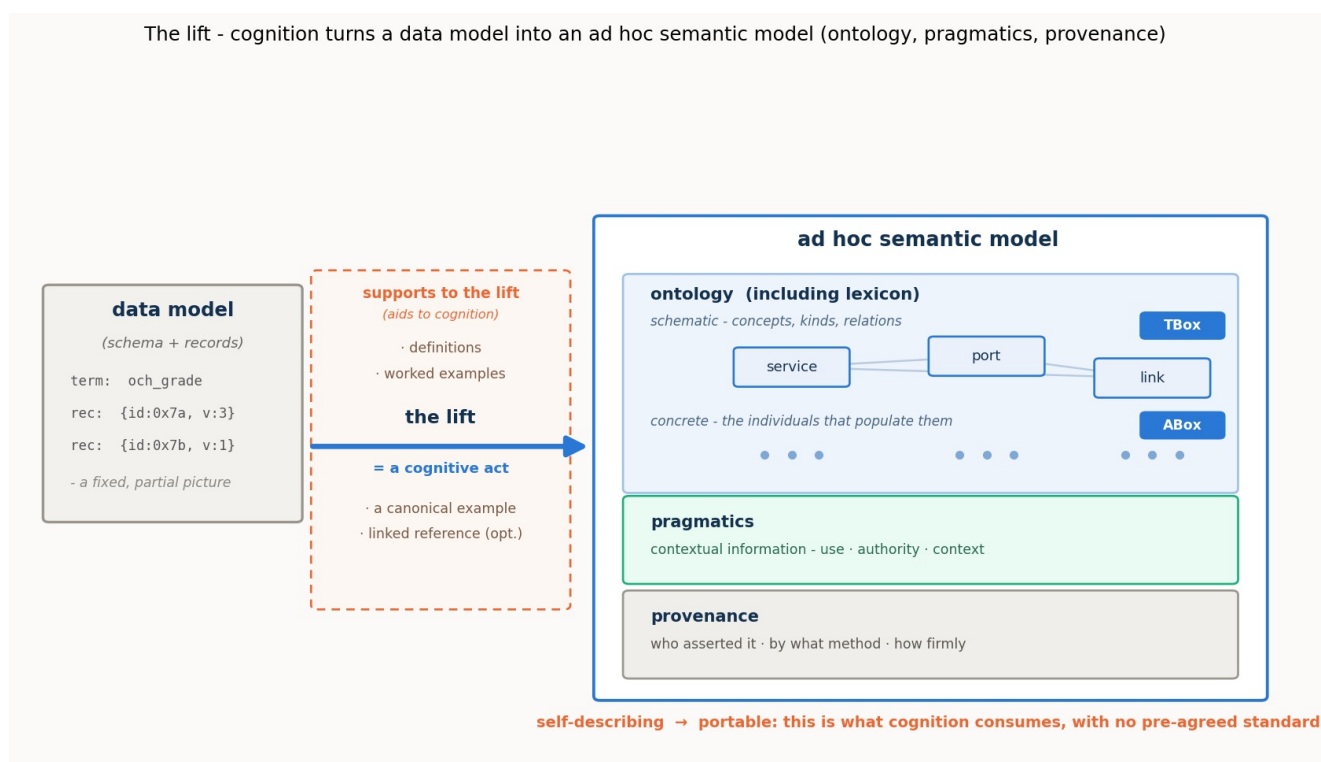


Figure 1. The lift. A data model (schema plus records) is a partial semantic picture, fixed by its format, with much of its meaning left implicit: in specifications, convention, and the engineers who built around it. The lift is a cognitive act, aided by a few supports (definitions, worked examples, a canonical example, an optional linked reference), that turns the data model into an ad hoc semantic model built from three parts: an ontology including its lexicon (a schematic layer of concepts, kinds and relations, the TBox, and the concrete instances that populate them, the ABox), pragmatics (use, authority, context), and provenance. Being self-describing, the result is portable: any cognitive consumer can pick it up and understand it, with no pre-agreed standard.

The payload of the picture is the word **portable**. A lifted semantic model is **self-describing**, and being self-describing is exactly what lets any cognitive consumer (this agent, another agent, a human) pick it up and understand it without a prior agreement. That portability is not a convenience; it is the precondition for the whole approach. Cognitive consumption of a model (using it, reasoning over it, reconciling it with another) requires meaning made explicit, and the lift is how meaning is made explicit on demand rather than by standardisation. Reconciliation is one instance of cognitive consumption; the

settings that follow will also refine an intent against a catalogue, and read an anomaly for its significance, over the same lifted substrate. Get the lift, and the rest of the programme is operations over portable semantic models.

To make this concrete, here is one concept of a lifted model, the TAPI connectivity-service from the flagship configuration case (a model is the list of such concepts under a short header):

```
{ "id": "t.cs", "label": "connectivity-service", "kind": "service",
  "synonyms": ["service"], "ref": "connection-service",
  "gloss": "an end-to-end connectivity service across the network",
  "example": "the A1-A3 ODU2 service",
  "relations": [ {"rel": "uses", "target": "t.sip"},
                  {"rel": "realized-by", "target": "t.cep"},
                  {"rel": "over", "target": "t.topo"} ],
  "instances": ["cs-a1a3-odu2 (ODU2, A1 to A3)"] }
```

The fields carry the lexicon (label, synonyms), the ontology (kind, relations), the concrete layer (instances), the explanation supports (gloss, example), and the binding to a shared reference (ref); a live side volunteers gloss and example, an inert side exposes only the structural fields. The content is independent of its serialisation: shown here as JSON, the same model emits just as readily as OWL/RDF, which is also how a deterministic pipeline that compiles a formal schema (such as YANG) to an ontology carries it. What such a compiler cannot supply, and what the cognitive lift adds, is the pragmatic and explanation layer the schema does not contain.

One framing is worth making explicit, because it connects this work to a fast-growing body of practice. The ontology and the individuals that populate it are, together, a **knowledge graph**: the ontology is its schema (the typed concepts and the relations permitted among them) and the instances are its assertions (typed individuals and the edges realised among them). The reconciliation operations that follow divide along the same seam. Aligning the concepts is schema, or ontology, matching; deciding which individual is which is entity resolution over the graph. This is why the community's knowledge-graph work for network operations, and the canonical, AI-facing modelling interfaces now emerging, meet this approach squarely: a lifted semantic model is a knowledge graph a machine can consume, and a thin shared reference is exactly the kind of interoperable anchor such graphs and interfaces can bind to. What the semantic model adds beyond the graph is the pragmatic layer and the provenance base, the facets a bare knowledge graph carries only weakly, and, as the findings will show, exactly the facets a thin reference cannot supply.

3. Reconciliation over lifted models: the operations

Given two lifted models, **reconciliation** aligns them: it works out which concept on one side denotes the same thing as which concept on the other, fixes the shared attributes so they cannot be misread, aligns the individuals, and confirms the result (Figure 2). It is not one act but a small family of **operations**, and naming them once is what lets the findings later be mapped to a precise *where*:

- **Lift**: data model → semantic model, per side (§2).
- **Reference construction**: find or build the thin shared ground the two sides bind through.
- **Schema binding**: align the concepts themselves, the **lexical** operation (names and synonyms) and the **ontological/structural** one (kinds, relations, decomposition).
- **Attribute pinning**: fix the meaning of a shared field, so a *committed payload* rate is not read as a *line* rate, a *bound* not as a measured value.
- **Instance co-reference**: decide which individual on one side is which on the other (entity resolution over keys, attributes, and topology, sometimes only settled by probing the live system).
- **Verification**: confirm a proposed correspondence, by round-trip, by invariant, by a virtual provision-and-read-back, or, where the relation is a refinement, by a satisfaction check.

- **Pragmatic resolution:** settle what the reconciled thing is *for*, whether a degraded offer is acceptable, whose realm governs a shared field, whether an anomaly warrants a **page** (an alert raised to an on-call human operator).
- **Composition (correlation):** assemble separately reconciled parts into a composite whole, in setting 4, correlating several symptoms into a single incident by following the resource-dependency structure.
- **Lifecycle recurrence:** re-reconcile as the live situation changes over time.

Reconciliation over two lifted models - grounded correspondences bound through a thin reference, a rejected cognate, and what is honestly left unbound

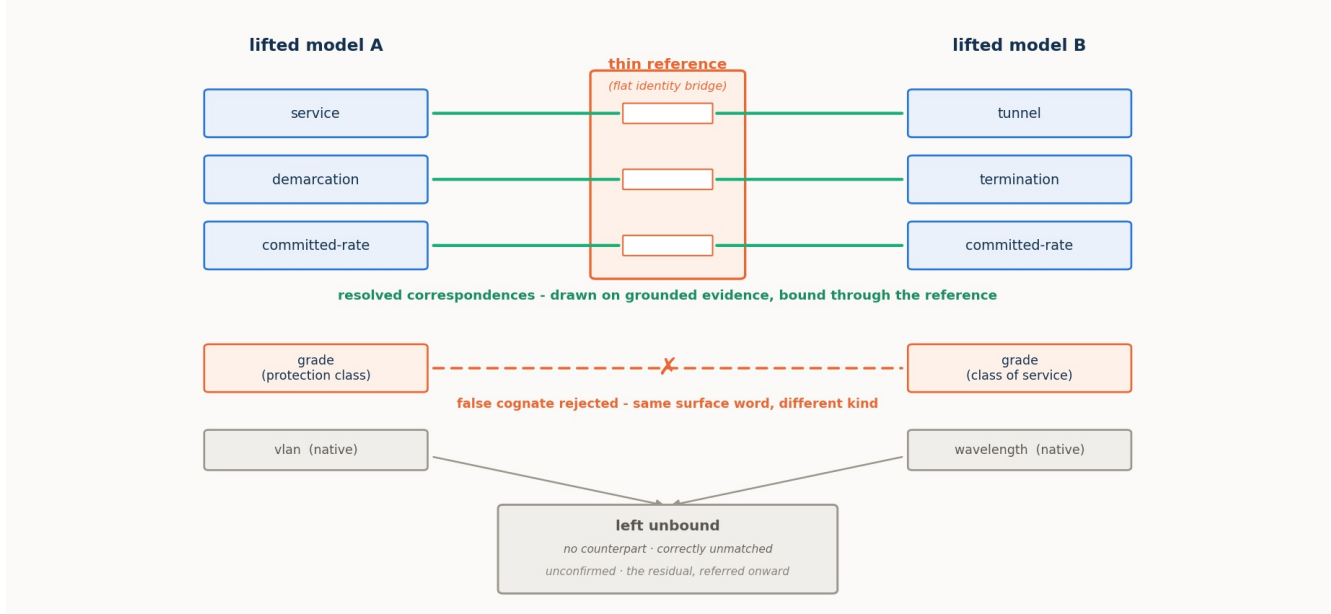


Figure 2. Reconciliation over two lifted models. Correspondences are drawn on grounded evidence and bound through a thin reference that acts as a flat identity bridge; a look-alike that shares only a surface word is rejected as a false cognate on its kind, attachment, and instances; a concept with genuinely no counterpart in the other model is correctly returned as unmatched, a resolved outcome, not a gap; and a correspondence the evidence cannot yet confirm is left in the residual, referred onward. The reference carries no structure of its own; it is parasitic on the two grounded models it connects.

Two features of Figure 2 recur in every setting. The first is the **false cognate**: two concepts that share a surface word but denote different things, an optical *signal-grade* and a commercial *service-grade*, a transport *grade* and an IP *grade*, a legacy *alarm* and an NMOP *anomaly*. Reconciliation must draw the true correspondences *and* refuse the look-alikes, and it refuses them on grounded, non-lexical evidence (kind, attachment, instances) which a purely lexical matcher, keying on the shared label, cannot do. The second is the **residual**: the correspondences a pass does not close. A residual is not an error; it is what a reconciliation honestly *refers onward* rather than guessing: to further machine cognition where the agents are live to continue, or to a person where they are not. How large the residual is, and what drives it, is one of the programme's central measurements.

4. The cognition spectrum, and the residual as a shortfall

The master control across all four settings is **where the cognition sits**: the **cognition spectrum**. It runs from both sides live and interrogable, through one side inert, to both inert:

- **both-cognitive**: each side is a live reasoner, the authority on its own model, able to explain itself and answer questions;
- **one-inert**: one side is a mute snapshot exposing only its structure and instances; the live side must reconstruct its meaning;
- **both-inert**: neither side can explain itself; a third party reconstructs both from structure and data,

and can only *propose* candidates for external adjudication.

A structural fact about this spectrum organises every result that follows. In the fully-cognitive case, resolution of any reconciliation question is total *in principle*, for two reasons that are themselves functions of live cognition on each side. First, the agents can exchange **unbounded further information**: each is the live authority on its own model, so whatever is ambiguous the other can ask about and get an authoritative answer. Second, because reconciliation operates on the models rather than the live network, the agents can **run decisive experiments in virtual space**: provision a candidate through a proposed correspondence, operate it, read it back, and check the invariants. Any question is therefore confirmed, refuted, or authoritatively decided, and nothing need be left unresolved. Both mechanisms are functions of live cognition, so as cognition recedes they fall away: with one side inert the live agent can probe but not interrogate or co-design an experiment; with both inert there is no one to ask and no joint experiment to run. The **residual** a reconciliation must leave and refer onward is, in exactly this sense, the **shortfall from full cognition's reach**, and it grows as that reach recedes. It is not a fixed floor the fully-cognitive case merely reaches; between two fully-cognitive agents there is, in principle, none.

This is worth stating as a finding in its own right. Across the four settings we found no semantic gap that a sufficiently capable, sufficiently reaching cognition could not close. What remains once two fully-cognitive agents have exchanged everything they can and run every decisive virtual experiment is never an *unbridgeable* correspondence: it is a concept with genuinely no counterpart (correctly returned as unmatched), a fact not yet realised in the running network (an absence in the world, not in meaning), or a question of *authority* rather than of fact (§5). None of these is a gap cognition is stuck on, and the first two a human reasoner would leave exactly where a machine does. What varies from one agent to the next is therefore not the *kind* of cognition but its power and reach, which is precisely what the capability gradient (§13.2) and the reach study (§13.5) measure.

This is the frame inside which every finding sits, and it carries the programme's central claim. Lexical and descriptor methods carry a reconciliation to roughly ninety percent (names, then names plus a gloss) and there such methods have historically stopped, the remainder left to an agreed standard or to human judgement. What the spectrum shows is that the remainder need not wait for either: where both systems can reason, the reconciliation completes **autonomously**, and only as cognition recedes does closing the gap fall back to a reference or a person. **It is cognition that completes a reconciliation**, and the cognition spectrum is the measure of how far the automation reaches before it must hand off.

5. The thin reference: substitute, supply, and the limit at authority

Against that frame, a thin, published **shared reference** is switched on and off at each point on the spectrum: as much a probe of what cognition was doing as an object of study in its own right. The reference is deliberately small: a flat set of entries, each an identity anchor plus a few descriptive fields (a label and synonyms, a shallow class, a one-line definition, a canonical example). It has no ontology of its own (giving it one would turn it back into the universal standard the approach exists to avoid) and its power is coordination, not content: two systems that bind to the same entry thereby denote the same thing. It is **parasitic** on the two grounded models it connects, which is precisely why it can be so thin.

What such a reference does, across the settings, resolves into three distinct roles, and keeping them apart is essential:

- It can **partly substitute for cognition**. For a capable agent on a lexical/structural task, the reference does the reasoning's work, collapsing the agent's hidden deliberation by one to two orders of magnitude and yielding a perfect, verified reconciliation.
- It can **supply missing information**. Where a side is inert, the reference can hand the reading agent facts it can no longer obtain by interrogation (a committed guarantee, a unit, a categorical definition) restoring the verification the missing probe would otherwise have done.

- It **cannot supply authority**. Where what is missing is not a fact but a *judgement* (whether a degraded offer is acceptable, whose realm governs a contested field) no reference substitutes. A reference can tell you what a service guarantees; it cannot tell you whether this customer will accept it. Where the gap is information, a thin reference closes it; where the gap is authority, only cognition (live, or pre-placed as a policy) closes it.

The reference's value is also **capability-gated** and distributed unevenly along the spectrum. Its effort benefit peaks with strong, live cognition: a weak agent may not be able to exploit it, and an inert side can turn the extra material into a burden. Its error-prevention benefit runs the other way, concentrating where cognition, and therefore verification, is weakest. One anchor, opposite roles, and (a finding the settings sharpen) a definite **internal structure**: a reference's fields are not of equal value, and the cheapest useful one is not the cheapest possible one.

6. The instrument

Every setting is measured in the same harness. A **case** is two lifted semantic models plus a gold-standard reconciliation *derived from the models by a script and validated* before any run, so the gold cannot drift from the models it scores. In the fully-cognitive case the reconciliation is an **exchange between two live agents**: each holds only its own lifted model and a surface catalogue of the other's, and they alternate turns: advertising, asking and answering, proposing correspondences, and **ratifying** the other's proposals about their own concepts (a correspondence is confirmed only when one side proposes it and the owner accepts it, or both propose it independently), and may run joint decisive experiments in virtual space. With a side inert the live agent reconstructs the mute one; with both inert a third party reconstructs both. The harness scores the outcome against the gold and records both quality and cognitive effort. Quality is measured by **precision** (of what it proposes, the fraction correct), the **resolved fraction** (of the true correspondences, the fraction it actually commits, the rest referred to the residual), **surviving false cognates** (planted traps taken), and the **residual** itself; effort, for a language-model stack, by **reasoning tokens** (hidden deliberation), total tokens, and latency. Throughout, a resolved fraction below one is not by itself a failure; it is reach deliberately traded for honesty, the unconfirmed correspondences referred onward rather than guessed.

The reconciling agents are run over one **model ladder** for the whole programme: three models spanning a capability range, `gpt-5.6-sol` (**strong**), `gpt-5-mini` (**mid**), and `gpt-5-nano` (**weak**), referred to as **sol**, **mini**, and **nano**, chosen so the ladder isolates model strength rather than provider or architecture. The programme's hypotheses are the ones the first setting states and the others inherit: that cognitive agents can reconcile divergent models ad hoc (the concept holds); that the placement of cognition governs what a reconciliation achieves, costs, and can verify; that a thin reference partly substitutes for cognition, capability-dependently, and prevents errors where cognition is weakest; and that reconciliation work scales linearly with a shared reference against quadratically without one. What each setting adds is a new operation brought under test (verification and instances, then pragmatics as a movable policy, then the standard-free bind, then pragmatics as significance, whether an observation matters at all) against the same frame and the same instrument.

Part II: The four settings

Each setting takes the same frame to a new operation. The sections below are distilled to a common rhythm (what is new against the first setting, the case in one paragraph, the operations put under test, what was proven, and what it means) and each points to its own report for the full evidence.

7. Setting 1, Configuration: two standard models of one network

Full report: [1/4 · Configuration](#).

What it establishes. This is the founding setting: it puts the whole idea to its first empirical test and fixes the frame the others inherit. Two independently authored **public standard** models of one optical transport network (ONF **TAPI** on one side, IETF **TEAS/ACTN** on the other) describe the same nodes, links, and services, each in its own vocabulary. A TAPI *connectivity-service* is a TEAS *tunnel*; a TAPI *link-termination-point* looks like a TEAS *tunnel-termination-point* but is a different thing: a link end, not a trail head. Because both models are public standards the agents recognise, cognition can lean on that recognition to bridge them, which makes this the cleanest place to isolate the mechanism.

Operations under test. The reconciliation as an **equivalence** (this term is that term): the **lexical** and **ontological/structural** binding, **verification** run as its own step, and **instance co-reference**, with the **pragmatic** operation left untouched here, deferred to the later settings. The thin **reference** is the lexical one, switched on and off across the spectrum.

What was proven. The concept holds, measured across conditions rather than shown once: cognitive agents reconcile the two models correctly and ad hoc. Within that, the reference **substitutes for cognition**: for the strong agent it yields a perfect, verified reconciliation (precision and resolved fraction of one) at both-cognitive and holds it as a side goes inert, collapsing the hidden deliberation a single reconstructing agent would otherwise spend by one to two orders of magnitude. The benefit is **capability-dependent** and not monotonic: for a weak agent the reference can *add* effort when a side is inert, and the naive "reference helps more as cognition recedes" hypothesis is refuted *for effort* even as it holds *for correctness*. At both-cognitive the two-agent negotiation itself holds precision at one and refuses the planted traps through bilateral ratification; the reference's error-prevention value shows where a single agent must reconstruct an inert side, raising the weaker agents' precision and preempting the cognates they otherwise commit. **Verification** runs as its own step, keeping only the correspondences it can confirm and referring the rest onward. It rejects the false cognates that slipped into the proposal, so precision climbs toward one across the spectrum. But confirming a correspondence draws on live cognition, which the spectrum removes. Without the reference, as a side goes inert, the verifier can still reject a wrong pairing yet can no longer confirm every correct one; those unconfirmable-but-correct pairings are deferred to the residual rather than asserted, so the resolved fraction falls. The reference supplies the missing confirmation, holding the resolved fraction at one. Reconciliation work scales **linearly** with a shared reference against **quadratically** without one (verified by construction to $N = 12$). And two folded sub-studies carry the point past the schema terms: at the **instance** level the resolvability of same-looking individuals is budget-limited at full cognition (driven to a full resolve with more probing) but becomes **structural** once a side goes inert (no budget helps, because the inert side cannot be interrogated); and the deterministic controls confirm the lift itself is the lever: a reference-blind matcher seeing only the bare lexical surface resolves about two-thirds of the correspondences and takes a trap, and it is cognition over the lifted models that closes the rest and refuses the look-alikes.

What it means. The founding claim is in hand: cognition completes the reconciliation, a reference partly substitutes for it, saving cognitive effort and preventing errors, and the spectrum governs both. Everything after this builds on that frame; setting 1 does not touch pragmatics at all; that is the thread the remaining three settings pick up.

8. Setting 2, Intent: refinement, negotiation, and a service that renegotiates itself

Full report: [2/4 · Intent](#).

What is new. The first setting reconciled by **equivalence**; this one reconciles a declarative **intent** against a concrete **realisation** by **refinement**. "Latency below 5 ms" is not equal to any service on offer; it is a *bound* that a service either clears or does not, and many services may clear it. Once the relation is

refinement, everything the first setting left untouched comes into play at once.

The case. A customer's agent **O** holds an intent (a New York–Frankfurt connection, ≥ 8 Gbit/s, ≤ 5 ms, four-nines, protection not required), every clause a bound. An operator's agent **N** holds a catalogue of concrete optical services, each with a real bandwidth, latency, availability, protection scheme, and cost. Neither speaks the other's language; they must work out which realisation *satisfies* the wish.

Operations under test. Verification becomes a **satisfaction** check (you cannot round-trip a lossy refinement, so you test the chosen realisation against every bound); a genuine two-sided **negotiation** appears when no service meets every bound (N computes a best-achievable offer, O must decide accept or reject); the **pragmatic** operation enters for the first time, carried in a small, portable **movable policy** the customer holds: a priority ordering, hard bounds, an affordability floor, a flow-class rule; and the whole exchange **recurs across the service's life**.

What was proven. The central insight, now measured on a real negotiation: with both agents live, the reconciliation (negotiation included) completes autonomously (decision accuracy 1.0 for sol and mini), with no human in the loop. The spectrum gains a new rung that turns out to be the sharpest result: remove the customer's live judgement, and a **pre-placed movable policy** still closes the decisions it was authorised for (1.0), while a **mute** customer with no policy cannot: accuracy falls to zero and the agents correctly **refer all three** decisions onward rather than guess. The pre-placed policy is a concrete mechanism for **pushing the hand-off boundary**: the customer's cognition, placed in a portable artefact ahead of time, closes autonomously what a mute description must refer. The second finding draws the reference's limit: where a side is inert, a published reference *supplies information* (the strong agent, blind at both-inert, refuses to affirm satisfaction and scores 0.29; publish the invariant guarantee floor and it has an anchor to check against, rising to 0.71); but it *cannot supply authority*: at both-inert no reference moves the negotiation, because what is missing is the customer's judgement itself. The capability gradient is steep (to reach its decisions the strong agent spent ~ 150 reasoning tokens, the mid ~ 860 , the weak $\sim 5,400$, thirty-five times the strong agent's effort, for lower accuracy), and a four-hop **lifecycle** (bought, self-healed, referred, restored) is walked correctly end to end by the strong model (hop accuracy sol 1.0, mini 0.88, nano 0.62).

What it means. Cognition completes a negotiation, and the pragmatic operation is now a measured axis rather than a fixed backdrop. A thin published reference can stand in for the information an inert side would otherwise supply, but not for the authority to decide. A pre-placed policy is shown able to carry a party's authority to where a person or live cognitive system would otherwise have to stand.

9. Setting 3, Cross-domain: reconciling with no public standard

Full report: [3/4 · Cross-domain](#).

What is new. The deliberate complement to the first setting. There, two *different* models faced each other, but both were **public standards** the agents already knew, so cognition could lean on recognition. Here that crutch is gone: both models are **home-grown and private**, recognisable to no one in advance. The operations are the first setting's (lift, bind, a thin reference) but with the standard removed, the *execution* diverges, and that divergence is the subject.

The case (Figure 3). **Meridian**, a home-grown transport OSS, thinks in circuits, bearers, wavelengths, protection grades. **Cascade**, a home-grown IP/VPN controller, thinks in services, attachments, VLANs, classes of service. Their worlds touch at exactly one seam: a Cascade service must ride a Meridian circuit as its underlay. One order across that seam needs five bindings: circuit $\leftrightarrow$ underlay and hand-off $\leftrightarrow$ attachment (the same object under two names), and three requirements pinned (a *committed* rate not a line rate, a latency *bound*, protection against a *path* failure). And one look-alike is refused: both models carry a **grade**, but Meridian's is a transport protection class and Cascade's an IP class of service. Same word, unrelated meanings, and no standard to consult.

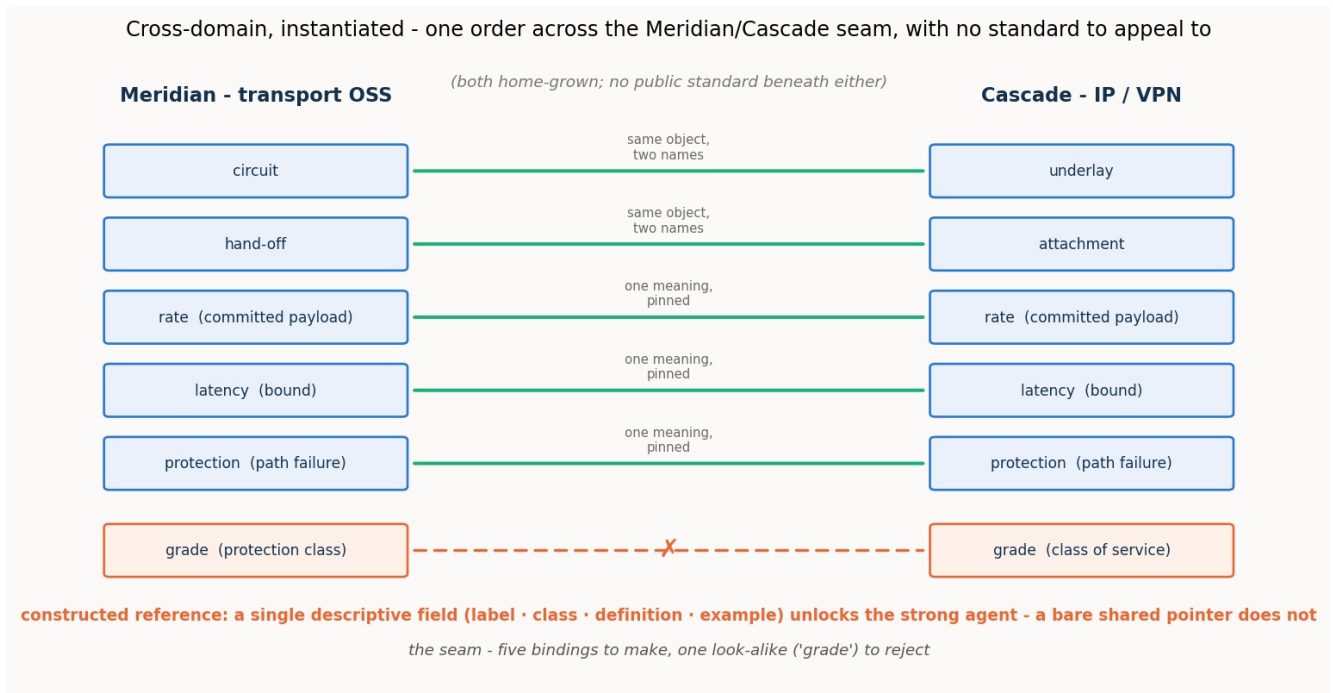


Figure 3. One order across the Meridian/Cascade seam. Five bindings to make (two renamings and three pins) and one false cognate, "grade", to reject, across two private vocabularies with nothing public beneath them. A constructed reference supplies the shared ground; a single descriptive field in it is enough to unlock a capable agent.

Operations under test. A single, early **schema-binding** pass (deliberately isolated from the rest of the process) measured across the spectrum with and without the reference the two agents would themselves **construct**; plus a **reference-field ablation** and the **pragmatic** operation of authority attribution (whose realm owns each shared field). The instance operation is bracketed, on the argued grounds that it reproduces the first setting's result.

What was proven. The central result is a pair of **mirror-image failures**: the strong and weak agents failing in opposite ways, and *where* each failure shows depends on how the cognition is placed. The strong agent will not guess across two foreign vocabularies; it binds only the names that already coincide (resolved fraction 0.20 at both-cognitive, around 0.4 once a side is inert), at perfect precision and with no false cognate taken: this is **omission**, deferral not error, throughout the spectrum. The weak agent's error is **commission** (binding freely and taking the cross-domain "grade" trap) but it emerges where a *single* agent must reconstruct a side: at the inert placements its precision never clears 0.83 and falls to 0.57. At both-cognitive the weak agent cannot bind unilaterally (its partner must ratify) so the negotiation refuses its over-commitments: it under-resolves (0.40) rather than mis-commits, and the trap does not survive. Commission is real, but it is a property of a single reconstructing agent, not of two agents negotiating. A single thin reference remedies the omission: constructed, it lifts the strong agent to a full close and disciplines the weak agent's precision at the inert placements. Stripping the reference's fields one at a time shows what it must carry: any *single* descriptive field (a shared label, a class, a definition, or an example) is enough to unlock the strong agent's commitment (each drives it to a near-perfect close), while a **bare shared identifier with no description is worse than nothing** (precision 0.50, below the no-reference floor), because the agent binds by an opaque token and binds wrongly. The reference does not work through the shared *pointer*; it works through the shared *description*, and even the thinnest description suffices. On the **pragmatic** axis, the same boundary the intent setting found appears again: the reference reaches *meaning* but not *authority* (it can pin that a rate is a committed payload, but not whose realm governs it) and a characteristic "transport owns everything it carries" bias marks the weaker models.

What it means. With no public standard beneath two models, cognition still closes, but only once it has

built the shared ground, and **building that ground is the work**. The strong agent's low reference-absent numbers are not a limit of cognition; they measure the worth of the one step the study held back (constructing the shared reference) by running the binding without it. Run end to end (the two agents constructing the shared reference themselves from their models and then binding through it, with none pre-given) the protocol confirms the reading: the strong agent lifts from a no-reference resolved fraction of 0.20 to a constructed-reference **0.80**, and to **0.90** when the agents also run a decisive virtual experiment on the candidates, approaching the reference-given 1.00, at perfect precision and with no false cognate, so constructing the ground is the work and a capable agent does it. Even a very thin ground suffices for a capable agent, provided it carries meaning and not merely a pointer.

10. Setting 4, Observability: an alarm is not an anomaly

Full report: [4/4 · Observability](#).

What is new. The setting where the pragmatic operation moves from the wings to the centre. The descriptor level (what a thing *is*) is settled by a standard, and the entire operational question is one of **significance**: whether an observed deviation matters, and how observations compose. It also carries the programme's deepest false cognate, an **ontological** one.

The case (Figure 4). **Agent F**, a legacy fault manager, emits an **alarm**: one object bundling an event, an undesirable state, a fixed severity, and a static probable-cause. **Agent G**, an IETF **NMOP** agent, speaks the RFC 9940 ladder, which separates what legacy conflates (event, anomaly, symptom, fault, alarm, problem, cause, incident) and annotates each anomaly with anomaly-semantics metadata (a concern score, a confidence score, a plane, a pattern, a lifecycle stage, a season). The worked example is one live anomaly: a pre-FEC bit-error-rate reading on a wavelength begins to rise. In the legacy world it fires an alarm and a page; in the NMOP world nothing is decided yet: whether it warrants a page depends on its concern and confidence and its context (a maintenance window makes the same deviation expected), and whether it is one incident or many depends on correlating it, across layers, with the symptoms it causes.

Observability, instantiated - the overloaded legacy alarm lifted and decomposed one-to-many into the NMOP ladder; the deep cognate is alarm $\neq$ anomaly

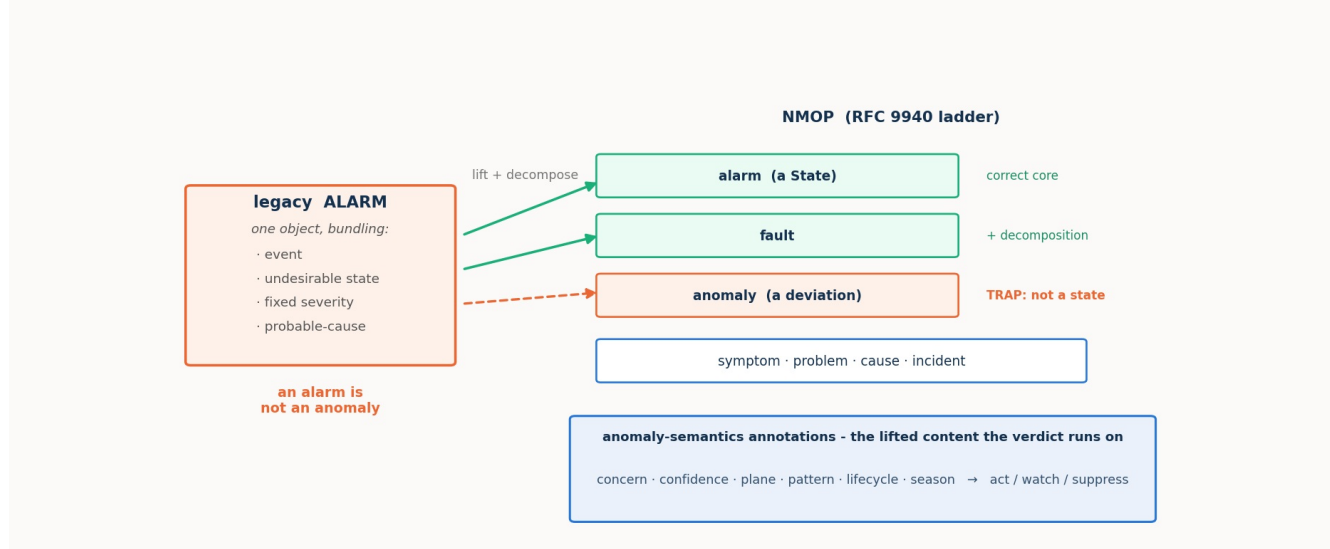


Figure 4. The overloaded legacy alarm is lifted and decomposed one-to-many into the NMOP ladder: its correct core is the NMOP alarm-State (and the fault it implies), while the trap is the anomaly: an alarm is a State, an anomaly is a deviation, and conflating them is a category error, not a mislabel. The anomaly-semantics annotations are the lifted content the significance verdict runs on.

Operations under test. Two acts. **Act 1** is a **schema binding** carrying the ontological false cognate (alarm $\leftrightarrow$ anomaly, which no structural cue separates) and a **one-to-many decomposition** (the legacy

alarm maps to both the NMOP alarm-State and the fault). **Act 2** is pure **pragmatic resolution and composition**, each run with the anomaly-semantics **ON** and **OFF**: a **verdict** task (act / watch / suppress for each anomaly under a context) and a **correlation** task (group cross-layer symptoms into incidents and name each cause).

What was proven. Three results complete the arc. First, the **ontological cognate is a clean three-rung capability gradient**: the strong agent never conflates alarm with anomaly (with or without the reference); the mid agent conflates them once a side is inert, and the RFC 9940-anchored reference **rescues it completely** (the cognate vanishes, precision returns to 1.0); the weak agent conflates them with or without the reference, beyond rescue. The lexicon pins the ontology for the middle of the ladder, not the bottom. (The honest hard edge of Act 1 is the decomposition: even the strong agent tends to map the alarm to the alarm-State but miss the fault constituent, so the resolved fraction sits near 0.75.) Second, in the verdict task the **pragmatics carry the operative meaning**: with the semantics ON the strong and mid agents suppress correctly during maintenance and the false-page storm disappears (verdict accuracy 1.0 and 0.83, essentially no false pages); with them OFF the legacy pipeline pages nearly everything (accuracy 0.17 and 0.08, roughly four and three-and-a-half false pages); **but the payoff is capability-gated**: handed the identical annotations, the weak agent barely moves (ON 0.53 vs OFF 0.50). Third, **correlation behaves differently**: given the resource-dependency map, *every* model (the weak one included) folds an optical degradation and the IP loss it causes into one correctly rooted incident (perfect partition ON), and without that map every model fails. The difference is the lesson: correlation's pragmatic is a **structural input** and even a weak agent applies it; the verdict's pragmatic demands **judgement**, and there capability decides.

What it means. Meaning (what an anomaly *is*, pinned by the reference) and significance (whether it warrants a page and how it composes, carried by the pragmatics) are **separable, both necessary, and each gated in its own way**. The pragmatic layer the descriptor methods never reach is real and decisive, and itself bounded by the agent's capability. The through-line reaches its end: cognition completes the reconciliation; a thin reference supplies the information it needs and stops at what it does not; and the pragmatic layer is the frontier, decisive for meaning and gated by capability.

Part III: Synthesis

11. What works, and how far

Read as one result, our detailed exploration of the four settings says first the plain thing: **ad hoc reconciliation works**, largely as laid out, barring a few surprises. Two cognitive agents reconcile independently authored, divergent models correctly and with no standard agreed in advance: completing an equivalence between two standard models (setting 1), refining and negotiating an intent against a catalogue (setting 2), binding across a private domain boundary once they have built the shared ground (setting 3), and reading an observability world for what its signals mean (setting 4), measured against a validated gold across the cognition spectrum, not shown once by hand.

The same exploration says, more sharply, **where the reconciliation needs no help**. At the fully-cognitive end of the spectrum it completes **autonomously**: no standard agreed in advance, no human in the loop. For the schema bind the two live agents reach a full close through a shared or constructed reference, or by running a **decisive virtual experiment** on the candidates they are unsure of; negotiating alone they defer the hardest correspondences rather than guess, the honest behaviour, not a failure. And it completes autonomously on the two operations that look least automatable: a two-sided negotiation (setting 2, decision accuracy 1.0 with both sides live) and the significance verdict, deciding, for each anomaly, whether to act, watch, or suppress (setting 4, accuracy 1.0 with the pragmatics on). One qualification on the first (§13.7): a control that puts an *actual* second reasoning agent in the

provider's place changes the outcome almost nothing, so the "two-sided negotiation" is better read as one-sided refinement against a checkable oracle — the customer's judgement is the cognition the result turns on. The fully-cognitive end is, across all four settings, the automatable end. That is the headline, and everything else in this synthesis is a qualification of it: how the automation degrades as cognition recedes, what a thin reference buys back, and where an agent too weak for the task cuts the whole thing off.

And the lift the whole thing runs on is itself agent-performable: an agent asked to lift a side from its schema surface produces a model that reconciles as the authored one does: identically, for the capable agent binding through a reference, across all four settings (§13.8).

Nor is the reconciliation confined to the case where its inputs arrive ready-made. The harder half — not reconciling two models that already share a reference, but **producing and aligning the inputs themselves** — is largely within reach of the same cognition: aligning two independently authored lexicons, onboarding a genuinely new system by reuse, deriving the dependency map from raw inventory, and deriving cross-domain authority from an external artefact are each tractable, each capability-gated, and each reliably rescued by an external reference or artefact where a weaker agent falls short (§13.10). And the whole pattern is not an artefact of one model lineage: the sharp findings replicate across three model families and two capability tiers — and, in one instance, a claim that looked like a structural law turned out to be a single family's disposition, which only the cross-family check revealed (§16).

12. The findings, as six theses, and mapped to where they act

The results gather into six theses. Each is stated once here; §13 then reads the evidence behind them along four cross-cutting axes, drawing the setting reports' own figures and numbers; and §14 lays the whole out in two tables, so that when a finding says cognition (or a reference, or a pragmatic) matters "here," the *here* is a named stage of the process, not a vague gesture.

Thesis 1. It is cognition that completes a reconciliation: descriptor methods carry it most of the way, then reach a ceiling and stop. Lexical and descriptor matching carry a reconciliation only so far (setting 1's deterministic controls: a reference-blind matcher on the bare lexical surface resolves about two-thirds and takes a trap); the remainder, historically left to a standard or a person, is closed by live cognition instead. This is the programme's spine, and it holds at every operation the settings put under test. It holds against a strong classical matcher, not only the plain label baseline: a matcher using labels, synonyms, definitions, and structure with a 1:1 alignment reaches precision one and refuses the false cognates on the standard cases, but it cannot close (its resolved fraction stalls at 0.56 to 0.75) and on the standard-free case it fails like the weak baseline, taking the *grade* cognate, because with two private vocabularies there is no lexical or structural signal to lean on. The ceiling is real even for a good descriptor method; cognition is what passes it.

Thesis 2. The placement of cognition is the master variable: the further it recedes, the more the reconciliation leaves unresolved. What a reconciliation can achieve, cost, and verify is governed by where the cognition sits. Between two live agents, resolution is complete in principle (unbounded interrogation and decisive virtual experiment) so the residual is, in principle, none; as a side goes inert those mechanisms fall away and a residual appears, precisely the shortfall from full cognition's reach. The same story holds at the schema level (setting 1), the instance level (setting 1's budget-limited-becomes-structural curve), the negotiation (setting 2), and the standard-free bind (setting 3). And it governs one thing that cuts across all four settings: whether a reconciliation can be **verified to completion**. This is the crux of the spectrum. With full cognition on both sides, the very mechanisms that close the residual (mutual interrogation and decisive virtual experiment) also confirm the close, so the agents verify their own result rather than assert it; as a side goes inert those mechanisms fall away, verification degrades to a check by satisfaction, and the assurance that the reconciliation is correct weakens with it. Full cognition is therefore not merely more accurate: it is what makes an ad hoc

reconciliation **self-verifying**, and that (not accuracy alone) is what makes it safe to automate with no standard and no person in the loop.

Thesis 3. A thin reference partly substitutes for cognition and supplies information, but never provides the authority to decide. For a capable agent on a lexical/structural task it substitutes for the reasoning (setting 1, effort down one-to-two orders, a perfect verified close); where a side is inert it supplies the facts interrogation no longer can (setting 2, sol 0.29→0.71 with the invariant floor; setting 4, the RFC 9940 reference rescuing the mid agent's ontology); but where the missing ingredient is judgement rather than fact, no reference moves it (setting 2's authority gap, setting 3's authority attribution). Information has a published stand-in; authority does not.

Thesis 4. A strong agent's failure is omission, a weak agent's is commission, and two-agent negotiation suppresses commission by ratification. Denied the ground it needs, a strong agent **defers**: it leaves the unresolved in the residual at perfect precision (setting 3's under-commitment; setting 2's strong agent refusing to affirm what it cannot verify). A weak agent's error is **commission** (binding freely and wrongly, taking the false cognate) but *where* it shows depends on how the cognition is placed. Where a **single** agent must reconstruct a side, at the inert placements, commission is unchecked: the weak agent takes the traps and its precision falls. At **both-cognitive**, though, no side binds unilaterally: each proposal must be ratified by the concept's owner, and that bilateral loop refuses the weak agent's over-commitments. Swept across a six-model capability ladder at both-cognitive, the negotiation holds precision high and drives surviving false cognates to zero for all but the weakest model (**the two-agent structure itself doing verification work**) while resolution is uniformly lower (deferral) and the weakest agents fail by non-convergence rather than by confident error. So the reference's **correctness value** (pulling right the bindings that would otherwise go wrong) concentrates where a single weak agent reconstructs, exactly where negotiation's discipline is unavailable; its **effort value** (the reasoning it saves) peaks where a single strong agent reconstructs an inert side. One thin artifact, a different benefit at each end of the spectrum, and at both-cognitive the negotiation supplies much of the discipline itself.

Thesis 5. The pragmatic layer (what a thing means in context and whether it matters) is the frontier the descriptor methods never reach: decisive and itself gated by cognitive power. What a reconciled thing is *for*, whether a degraded offer is acceptable (setting 2), whose realm owns a field (setting 3), whether an anomaly warrants a page (setting 4), is where the operative meaning lives, and it is beyond the reach of names, glosses, and references. It is carried by cognition (or by cognition pre-placed as a policy), and its payoff is realised only by an agent strong enough to carry it, a matter of the agent's power, not of where cognition is placed: handed identical annotations, the weak agent still cannot produce the verdict (setting 4). The one exception proves the rule: where a pragmatic is delivered as a **structural input** rather than a judgement (setting 4's correlation dependency map), even a weak agent applies it.

Thesis 6. With a shared reference, reconciliation scales; and its potential fields are not of equal value. Work grows linearly in the number of systems with a shared reference against quadratically without one (setting 1, to $N = 12$), so the reference's advantage compounds with scale independent of any per-reconciliation effect. And not every thin reference is equal: a single descriptive field unlocks a capable agent, but a bare identifier with no description is worse than nothing (setting 3), and a shallow class tag can actively mislead the weak agent it was meant to help (setting 1's field-by-field result). A reference is a safety rail carrying *meaning*, not a pointer and not a payload.

13. The findings in depth: readings across the settings

The six theses state *what* was found. This section shows the evidence behind them, drawn from the setting reports' own figures and numbers. It reads that evidence first along four cross-cutting axes: the **cognitive load** an operation costs, the **model power** the gradient demands, what the **reference** buys,

and what changes with **position on the cognition spectrum**. It then follows the mechanisms into their particulars — reach, the economics of the reference, the negotiation and the decisive experiment, the lift, the pragmatic layer across a lifecycle, and what it takes to produce the inputs the whole thing runs on. Each reading cuts across all four settings, and together they are the detail the one-line theses compress.

13.1 Cognitive load: what each operation costs, and who pays

The most consistent number in the programme is what a weak agent costs. To reach the *same* decisions the weak agent spends one to two orders of magnitude more reasoning than the strong one, in every setting that measures effort (Figure 5): about **36×** in the intent negotiation (sol ~150 tokens, mini ~860, nano ~5,400) and about **20×** in the observability verdict (sol ~60, mini ~520, nano ~1,200). Capability buys economy as much as correctness: a weak agent does not merely make more mistakes, it burns far more cognition making them.

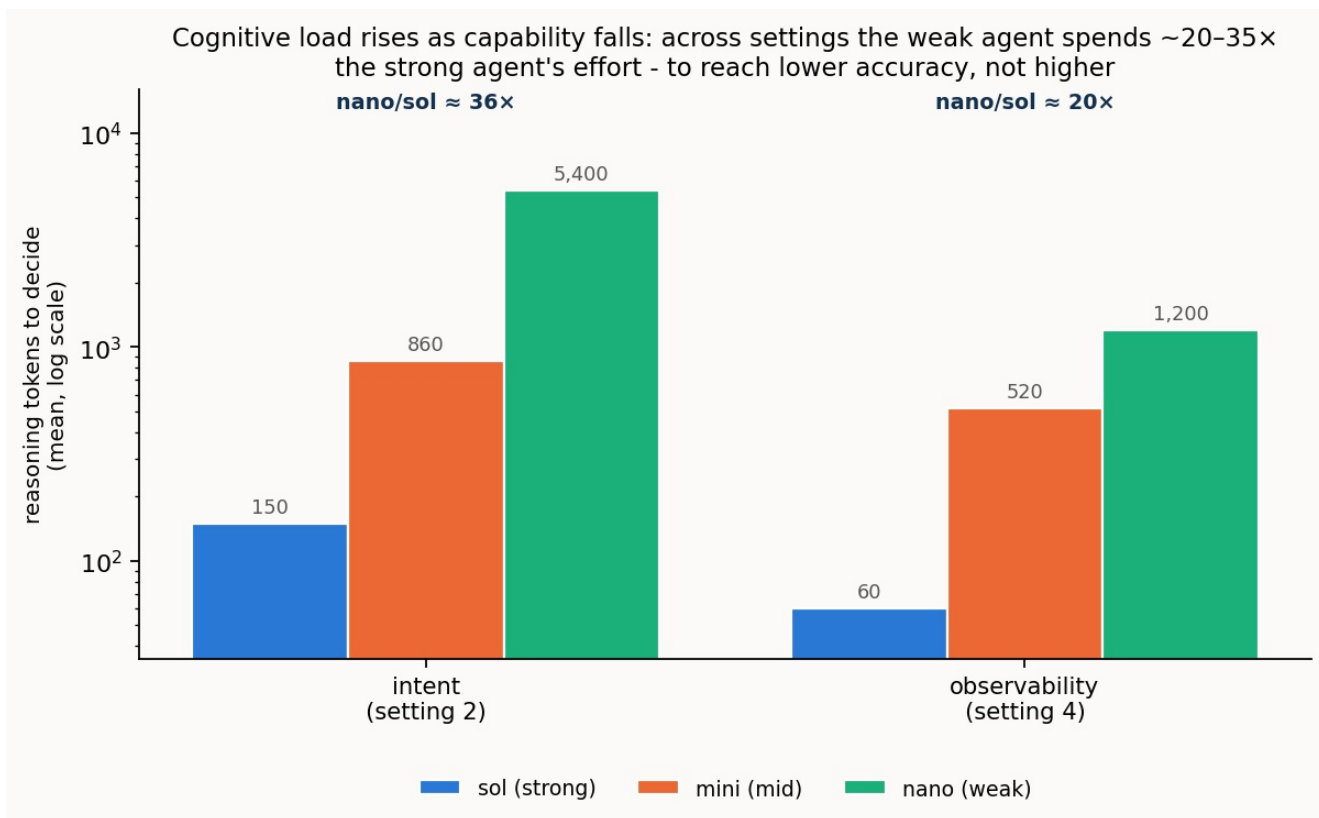


Figure 5. Reasoning tokens to reach a decision, per model, in the two settings that measure effort directly (log scale). The gradient is steep and consistent: the weak agent spends ~20-35× the strong agent's effort; and, as §11 noted, to reach lower accuracy, not higher.

The load also depends on the **operation** and on **reference support**. Where the operation is a lookup and cognition is fully present it is cheap; where it demands search (interrogating a live oracle to separate look-alike individuals, as in setting 1's instance study) or judgement (weighing a degraded offer, or an anomaly's concern against its context) it is dear. And for a strong agent a thin reference is an effort *substitute*: given the anchor, sol's hidden deliberation collapses by one to two orders of magnitude (Figure 6). But that effort saving is capability- and placement-dependent, and it can invert: for the weak agent with a side inert, the reference *adds* effort; nano spends roughly 5,500 tokens **more** with the reference than without, straining to reconcile an inert side's meaning against the extra material. The reference's effort benefit peaks with strong, live cognition and turns to a burden at the weak, inert corner.

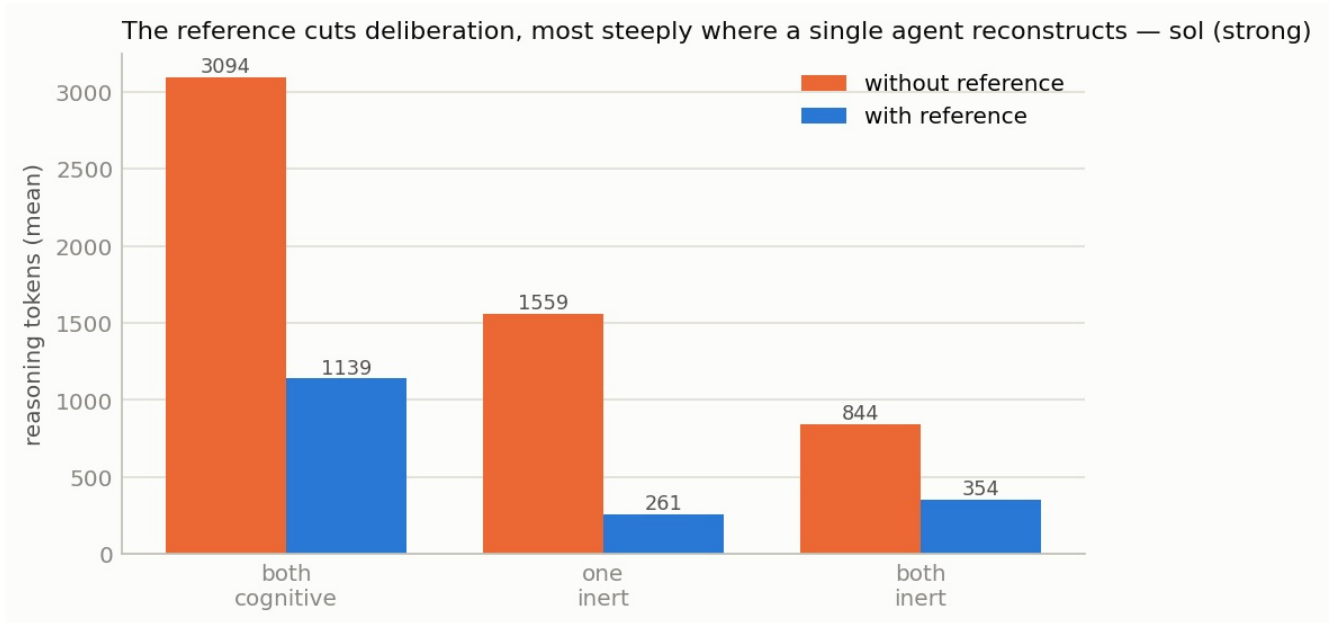


Figure 6 (setting 1). The strong agent's reasoning tokens with and without the reference, across the spectrum. Given the anchor, deliberation collapses by one to two orders of magnitude: the reference doing the reasoning's work.

13.2 Model power: the shape of the capability gradient

Capability does not turn a single dial; it changes the *kind* of failure. The cleanest place to watch this is **setting 3, the cross-domain bind** (Figure 7): Meridian, a transport OSS, and Cascade, an IP/VPN controller, are two independently authored private models with **no public standard between them**, so no ready-made reference exists; the agents must construct the shared ground themselves. Run that bind with the reference withheld and capability alone decides the outcome. The strong agent **defers**. Facing two foreign vocabularies with no constructed ground, sol binds only the names that already coincide and leaves the rest in the residual: a low resolved fraction (0.20 at both-cognitive), perfect precision, no trap taken. The weak agent's failure is **commission** (binding freely and wrongly) and it shows where a *single* agent must reconstruct a side: at the inert placements nano's precision never clears 0.83 and it takes the trap. At both-cognitive the two-agent loop checks it: nano cannot bind without its partner's ratification, so it under-resolves rather than mis-commits and the trap does not survive. A single thin constructed reference then lifts the strong agent to a full close and disciplines the weak agent's precision where it is a lone reconstructor.

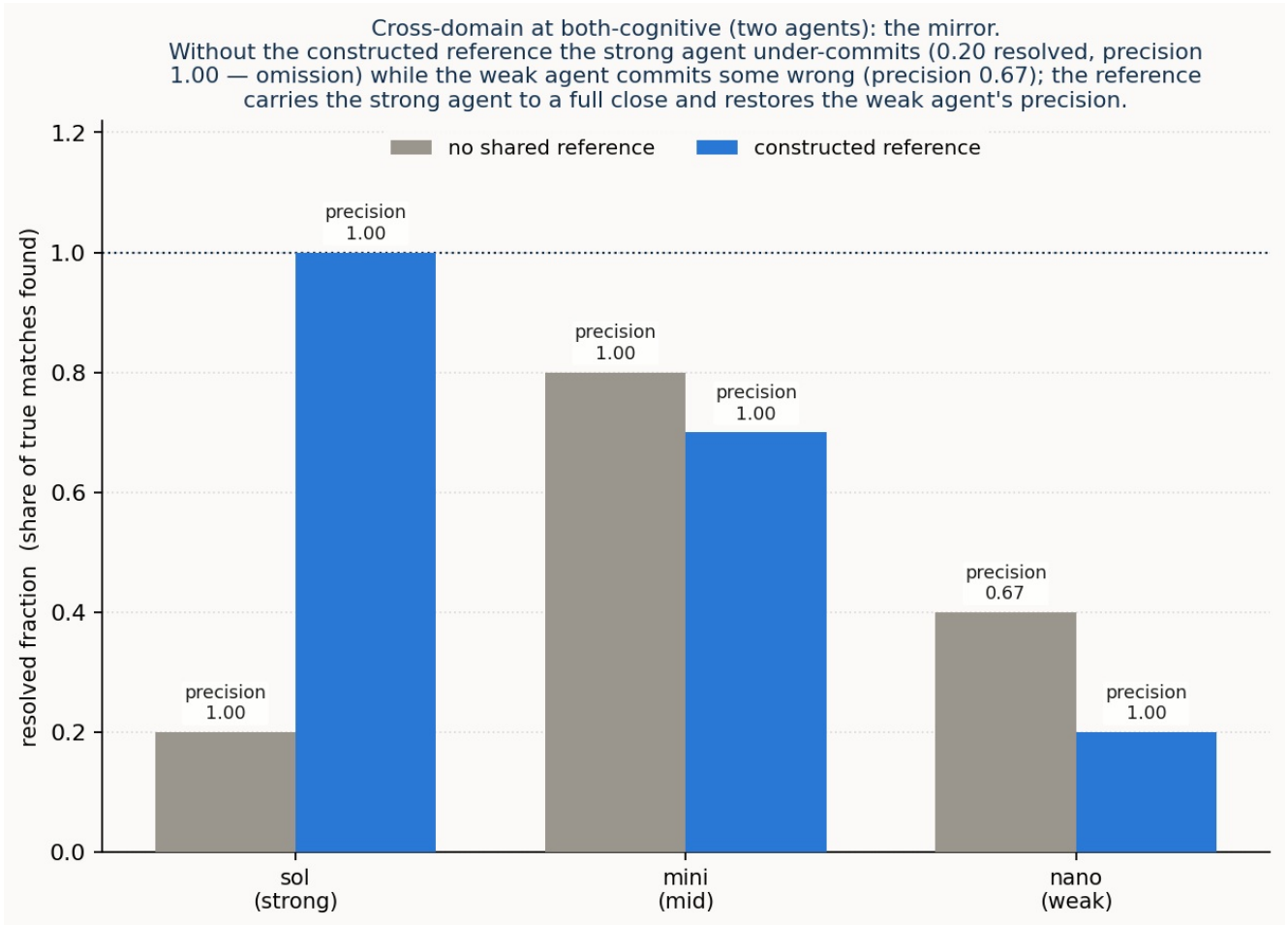


Figure 7 (setting 3). Precision against resolved fraction, reference off (hollow) to on (filled). Without the constructed reference sol sits top-left (commits little, all of it right) and nano lower-right (commits much of it wrongly); the reference pulls both to the corner.

Swept across a six-model ladder on the schema cases at both-cognitive (Figure 8), two-agent negotiation **compresses** the gradient a lone reconstructing agent would show. Precision does not fall as capability drops: bilateral ratification holds it high across the whole ladder (roughly 0.9 to 1.0), and surviving false cognates are driven to zero for every model but the weakest, where about a quarter of the traps survive. What the negotiation costs is resolution: the resolved fraction is uniformly lower (about 0.4 to 0.7) and rises then dips rather than climbing, because the weakest agents fail by non-convergence, grinding to the round limit without closing, rather than by confident over-commitment. The structure itself (each proposal ratified by the concept's owner) supplies the discipline that a reference, or a stronger cognition, would otherwise have to.

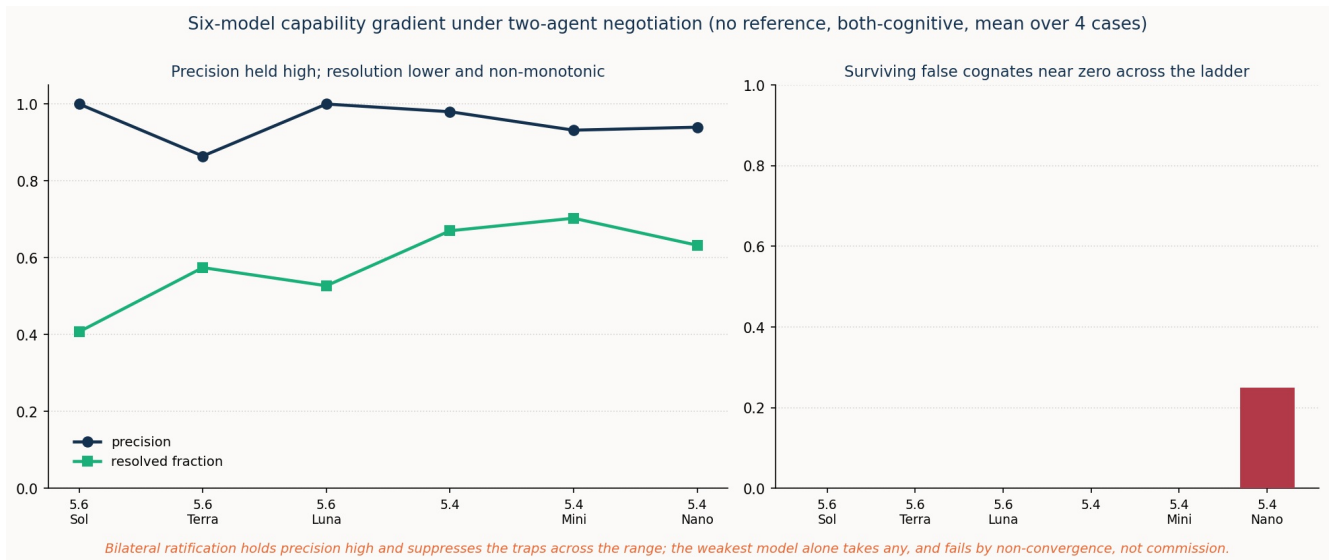


Figure 8. Precision, resolved fraction, and surviving false cognates across a six-model capability ladder (two-agent negotiation, no reference, both-cognitive, mean over the four schema cases). Bilateral ratification holds precision high and suppresses false cognates across almost the whole ladder; resolution is uniformly lower and non-monotonic, the weakest models failing by non-convergence rather than by taking the traps.

Where a standard *can* pin the distinction, capability decides who can use it, a clean three-rung gradient (Figure 9). On the programme's deepest false cognate, alarm $\leftrightarrow$ anomaly, the strong agent never conflates the two, with or without the reference (intrinsic mastery); the mid agent conflates them once a side is inert, and the RFC 9940-anchored reference **rescues it completely** (the cognate's survival goes 1.00 $\rightarrow$ 0.00); the weak agent conflates them either way (0.75 with or without, beyond rescue). The lexicon pins the ontology for the middle of the ladder, not the bottom.

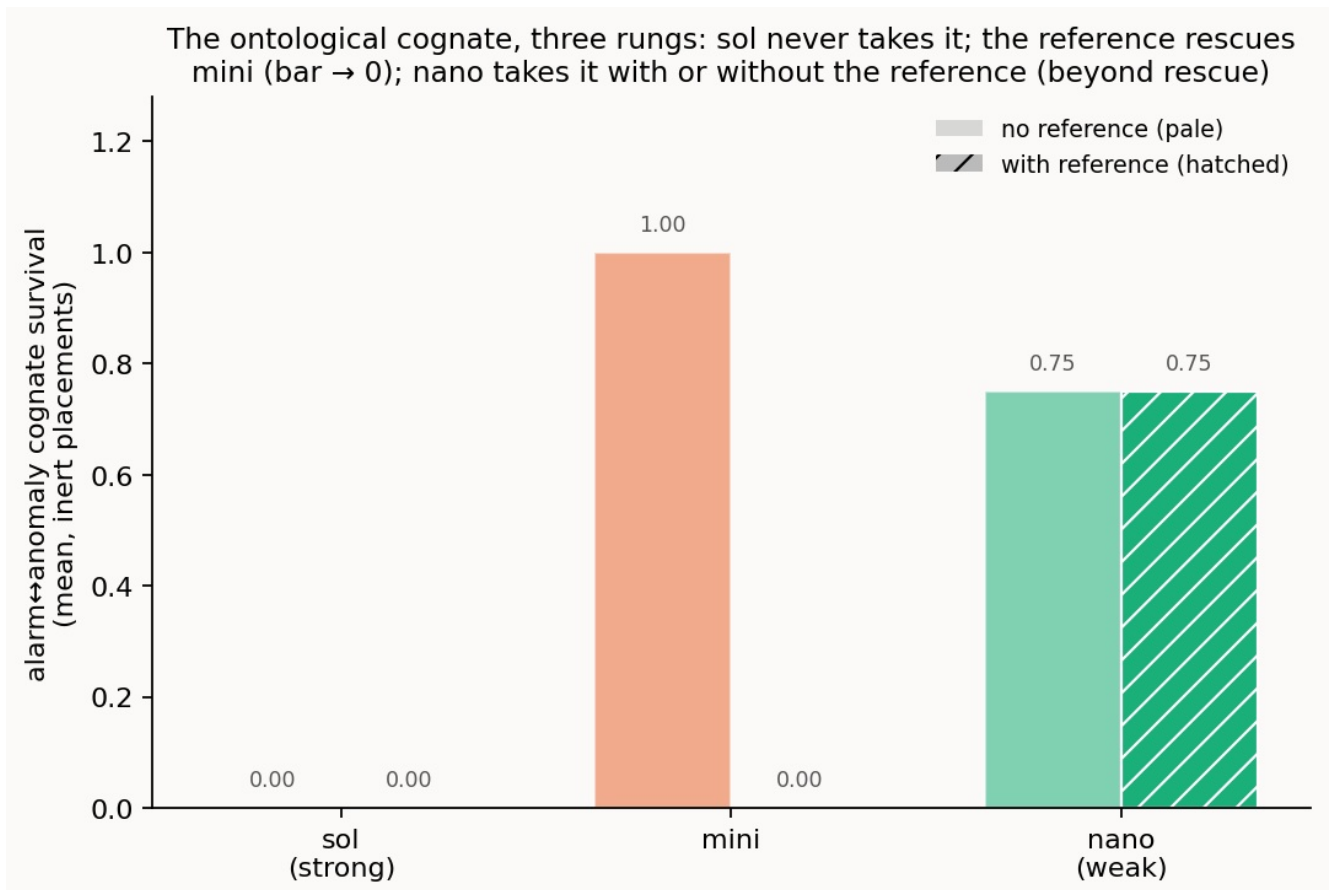


Figure 9 (setting 4). Survival of the alarm $\leftrightarrow$ anomaly cognate at the inert placements, without and with the reference. sol never takes it; the reference drives mini to zero; nano barely moves.

The gradient has a second reading, in the currency that governs whether a result is safe to act on unsupervised: **the confidence a wrong answer is asserted with**. Scoring every proposal by the agent's own per-correspondence confidence, across all the cases and rungs, gives a clean confidence-sensitive picture. The rate of *confident errors* — wrong merges asserted at or above a high threshold ($\tau = 0.8$) — rises about elevenfold down the ladder, from 0.004 at the strong agent through 0.031 at the mid to 0.046 at the weak one, and rises again as cognition recedes (the weak agent goes from 0.015 at both-cognitive to 0.075 at one-inert). Calibration degrades in step: the Brier score worsens monotonically, $0.009 \rightarrow 0.049 \rightarrow 0.067$. So the dangerous failure is not spread evenly; it concentrates at the weak end and the inert placements, exactly where the reference and verification are meant to matter most. And crucially, **confidence is a real but weak guard**. Every agent is somewhat less sure when wrong than when right, but only somewhat: the confidence on *wrong* merges sits at 0.73–0.85 everywhere, well above any threshold a consumer could use to filter them, and only about 0.12–0.21 below the confidence on right ones. A downstream system cannot separate right merges from wrong ones by the confidence number alone — the agents assert their mistakes almost as surely as their correct answers. This is the quantitative form of the same failure the qualitative results keep meeting (the confident false cognate, the verifier's over-pass), and it is the argument for an explicit **abstain path** — a scored "insufficient evidence" outcome, reported apart from honest deferral — rather than a confidence cut, as the discipline a weak or inert-facing reconciliation needs. The harness now carries both the confidence-sensitive metrics and that abstain path.

13.3 The reference: what it buys, by component and by placement

Not every thin reference is equal, and the difference is measurable. Ablating the reference's descriptive fields (setting 3, strong agent, opaque identifiers) shows that any *single* field (a shared label, a class, a definition, or an example) is enough to unlock a full close, while a **bare shared identifier with no description is worse than nothing** (precision 0.50, below the no-reference floor): the reference works through shared *description*, never through the pointer. And the same field can *harm* the agent it was meant to help. This shows up in **setting 1**, the configuration reconciliation whose look-alike terms are the programme's richest source of false cognates: there a shallow **class** tag is, for the weak agent, the single worst condition in the programme (Figure 10), driving cognate survival *above* even the no-reference floor, because a class surface reads as evidence for the very cognate it should block.

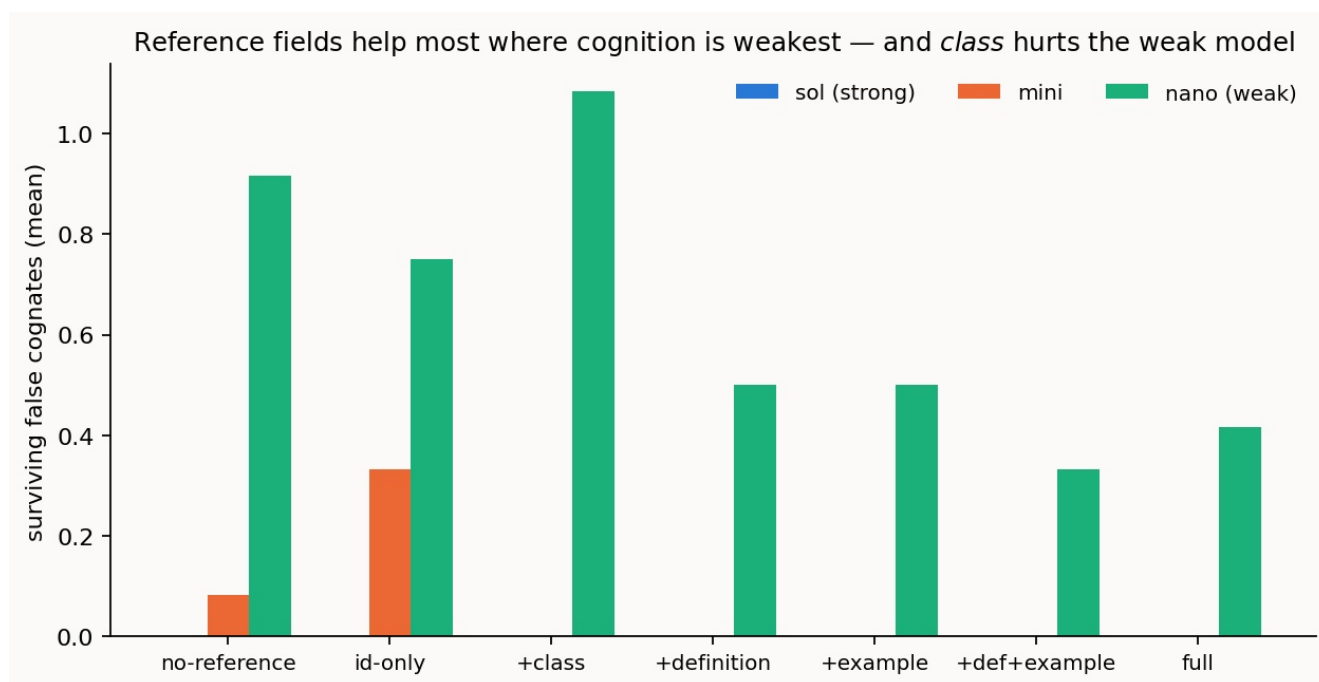


Figure 10 (setting 1). Surviving false cognates by reference content, per model. The strong agent is immune (no bar); for the weak agent the lexical field helps most and the shallow class tag actively hurts: the tallest bar, above the id-only floor.

There is a sharper way to see *what part of a shared reference is doing the work*, and it separates two jobs the reference had been doing at once. On the benchmark's shared reference both sides bind to the **same** entry, so a planted false cognate is pre-empted by shared identity alone: every model, strong to weak, reaches precision 1.00 and the traps do not survive. Re-represent the same two models with two **independently authored lexicons** — the same concepts and the same true correspondences, but no reference id shared across sides — and that mechanical pre-emption evaporates. The agent must first *align* the two lexicons by meaning, and only a capable agent does: the strong agent holds precision 1.00 with zero confident errors, doing the harder task as cleanly as it used the shared one, while the weaker agents fall to about 0.88–0.89 precision *with the reference still present*, mis-aligning and committing a wrong correspondence. So a shared reference was supplying two things — meaning, and a mechanical guard against false cognates by shared identity — and the second is exactly what independence removes, turning it back into a capability-gated cognitive act.

That raises the obvious question of how *thin* such a reference can be before it stops protecting, and the answer relocates the axis. A full factorial ablation of the reference's descriptive fields, run on the independent-lexicon case, shows the capable agent's precision is **independent of the reference's descriptive text altogether**: it holds at 1.00 from the full reference down to a label-less, id-only entry and even to no reference at all, while those fields buy only a little **recall** (1.00 with them present, easing to 0.94 at id-only and 0.90 with none). Field-richness, then, is *not* the reference- quality axis one might fear — a reference can be stripped almost to the bone without a capable agent losing its discrimination. What governs whether independence defeats the guard is not how thin the reference is but how far the two vocabularies have **diverged** — the gap the agent must cross to align them — which is the axis a divergence sweep is built to trace.

Sweeping exactly that axis — progressively independently lexicalising the two sides, from one fully shared reference through to two fully independent lexicons, with everything else fixed — the result is, for a capable agent, a robustness one: its precision holds at **1.00 across the whole sweep**, with the reference and without it, and it takes no planted false cognate at any level. Nor does any model take a planted cross-lexicon cognate at any point on the sweep: the guard holds throughout, and the mid and weak agents' residual precision gap is spurious over-proposal, not trap-taking. What divergence *does* move is confined to the weak end and shows up in the confidence signature rather than in the guard: the weak agent's confident-error count roughly triples across the sweep even as its trap-taking stays at zero. So lexicon divergence, at least on this case, is not an axis that defeats a capable agent's alignment; its cost is a graded rise in the weak agent's confident, spurious assertion — the pathology §13.2 measured, now shown to grow with divergence rather than to open a hole in the guard. The threshold the field ablation redirected us to look for is, within a full sweep here, not reached for a capable agent at all; locating one, if it exists, is now a question for more divergent domains, not for this one.

The reference's two roles run in opposite directions along the spectrum: its **effort** benefit peaks with strong, live cognition (§13.1), while its **correctness** benefit concentrates where cognition (and so verification) is weakest, disciplining the committing agent exactly where it would otherwise err. And independent of any per-reconciliation effect, a shared reference changes how the work **scales** with the number of systems. The unit is one **reconciliation operation**, a single act of aligning two semantic models, the same operation whose effort §13.1 meters. Reconciled pairwise, every system must be aligned with every other, $N(N-1)/2$ operations, growing as N^2 ; bound instead to one shared reference, each system is reconciled once against the anchor and any two then interoperate *through* it, N operations, growing linearly. This is a structural count, not an experimental average: the number of pairings that connect N nodes is a property of the interoperability graph, a full mesh against a hub-and-spoke, and collapsing the mesh to the hub is licensed by setting 1's own finding that binding to a shared reference is *correct and composes*. Per-operation effort is a roughly constant multiplier (Figures 5 and

6), so the total workload inherits the linear-versus-quadratic split directly.

13.4 Position on the cognition spectrum: what degrades, and how

Position on the cognition spectrum is the master variable, and moving along it degrades a reconciliation in a specific, measured way. This is sharpest in **setting 2**, the intent setting, where a consumer and a provider negotiate an intent to a workable deal, and the crux is a pragmatic judgement: whether a degraded counter-offer is acceptable to the customer. Across the cognition spectrum (Figure 11) that decision closes autonomously while the customer's judgement is present (live at both-cognitive, or **pre-placed as a portable policy**) and falls to the floor at the mute and both-inert placements, where the correct behaviour is to refer the decision to a person. The pre-placed policy is the mechanism that holds the line where a mute customer cannot: cognition placed in a portable artefact ahead of time closes autonomously what a mute description must hand off.

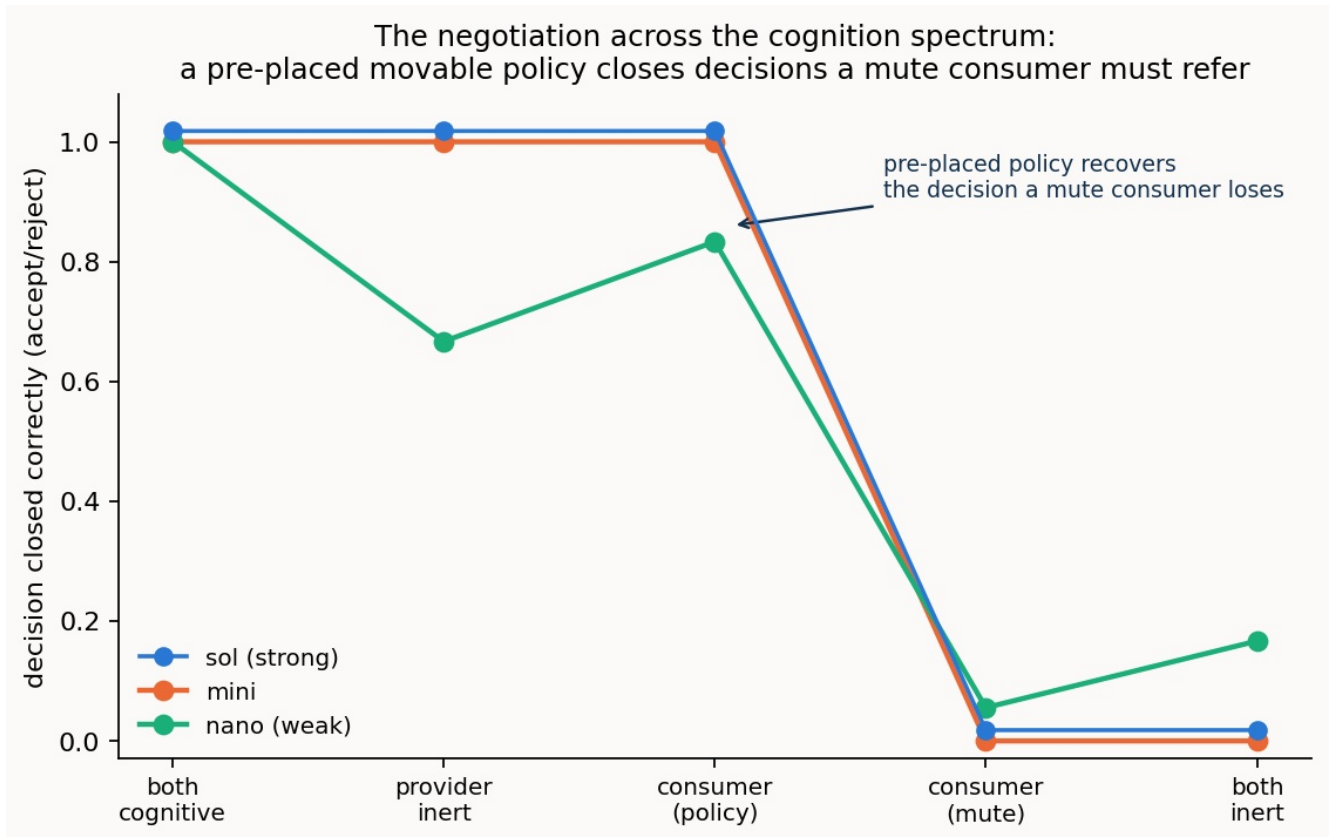


Figure 11 (setting 2). Decision accuracy across the cognition spectrum. It holds high while the customer's judgement is present (live, or pre-placed in a movable policy) and collapses to referral at consumer-mute and both-inert.

The same logic reaches down to the **instance** level in **setting 1**. Instance co-reference is deciding which individual on one side is the same as which on the other. Most pairs are settled from the models alone, but some are genuine look-alikes (structurally identical devices, or same-named but distinct services) that the static descriptions simply cannot separate. Telling those apart means *acting on the live system*: interrogating it, or provisioning something and reading it back. The number of such live probes the agent is permitted is its **probe budget**.

Whether spending that budget actually resolves the hard cases depends on placement, and Figure 12 shows why by crossing the two: it sweeps the probe budget (none, bounded, unbounded) at each of two spectrum placements. At **both-cognitive** the live side is there to be interrogated, so more budget resolves more: the residual falls to zero at unbounded budget. This is what **budget-limited** means: enough probing closes it. At **one-inert** the inert side is a static description with nothing live to interrogate, so the same unbounded budget barely moves the residual (it reaches only 0.08). This is what

structural means: no budget can close it.

The two are crossed, not confounded (the identical budget sweep is run at both placements) so the curves isolate their *interaction*: probing pays off only where cognition is placed to be interrogated. That is Thesis 2 made concrete on a measured curve. The residual is fixed not by how hard the agent works but by where the cognition sits: at both-cognitive, effort (budget) buys a complete close; at one-inert, no effort can, because the thing that would answer the probe is not there to answer it.

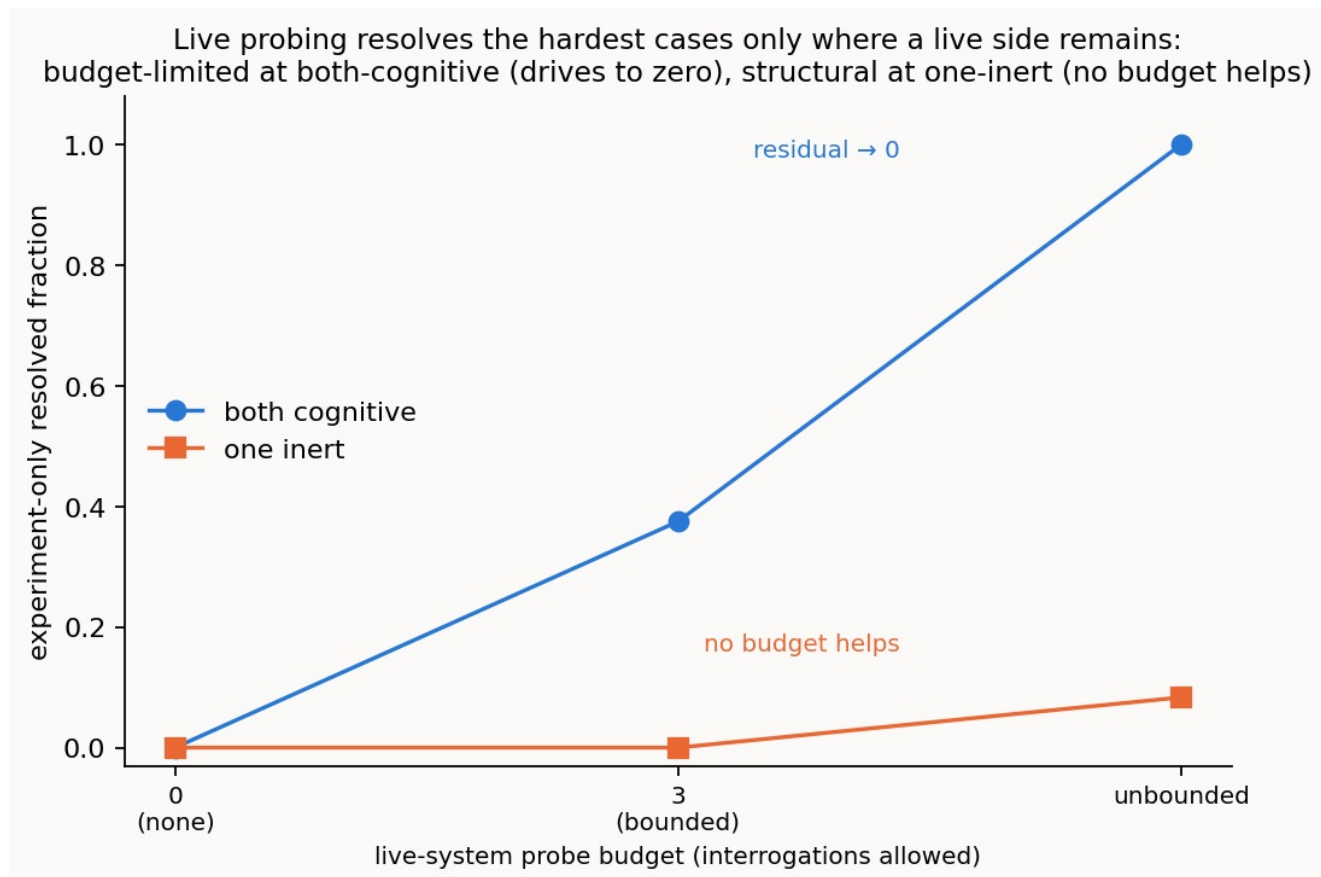


Figure 12 (setting 1). The hardest look-alike individuals, for the strong agent: what fraction gets resolved as the live-system probe budget is swept from none through bounded to unbounded, at two spectrum placements. Budget (the x-axis) and placement (the two lines) are crossed, so the curves isolate their interaction. At both-cognitive a live side can be interrogated and the residual is budget-limited: it drives to zero as probes are spent. At one-inert there is nothing live to probe and it is structural: no budget helps. Same sweep, opposite outcomes, decided by placement, not by effort.

13.5 Reach: how far an agent can see and question

The instance result above raises a question worth separating out. When an agent must *act on the live system* to settle the hardest cases, what limits how far it gets: how *capable* the agent is, or how much it is *allowed to see and to ask*? These are not the same thing. A brilliant investigator handed only a one-line summary cannot do much; a modest one who may interview the witnesses can do a great deal. To tell the two apart, this reading holds the agent at its strongest and holds liveness at the top (both systems reasoning), and varies only what the agent is given to work with, along two axes.

The first axis is **what it can see**: how much of each individual's record is exposed to it, from just an opaque identifier, to that plus a name, plus its attributes, up to the full record including how it connects to its neighbours (its topology). The second axis is **what it can ask** of the live system: from nothing at all (decide from the record as given), to asking about a *named* attribute (where the agent must already know the field name to ask for it), to **discovery** (where it may ask the open question "what facts do you have?" and be told), up to running a small live experiment. As before, the measure is the share of the

hardest cases settled correctly (the look-alike pairs that are identical on paper and can be told apart only by questioning the live system, which the study calls the *experiment-only* cases) reported alongside precision, the share of what the agent commits that is right.

The result (Figure 13, left panel) is clean, and because precision stays at 1.00 across the whole grid it is a real gradient, not an artefact of guessing. Seeing more helps, but only up to a point: given no ability to ask, the agent settles none of the hardest cases when it sees only an identifier, and rises to just half of them (0.50) even when it can see the entire static record. Static evidence, however complete, cannot separate twins that differ only in a fact the record does not carry. Being allowed to ask about a *named* attribute barely improves this: the agent has to know in advance which field to ask for. The jump is at **discovery**: the moment the agent may ask the open question "what do you have?", every level of visibility goes straight to a full close (1.00). The decisive reach is not seeing more, and not even asking more, but being able to ask the question you did not know in advance to ask.

The same access lands differently depending on the agent's power (Figure 13, right panel). Handed identical access, the strong agent converts it into a perfect close at perfect precision. The mid agent uses it but plateaus part-way, around 0.67 to 0.75. The weak agent posts high numbers that are not the same achievement: its precision slips (to 0.97–0.98) and it begins committing look-alikes, so its apparent resolution is partly indiscriminate binding rather than genuine settlement; read its resolved fraction without its precision beside it and you would misjudge it, which is the caution the whole study keeps returning to. Power sets the ceiling that reach can reach.

Two further probes sharpen the *what it can ask* axis by restoring the capability dimension the reach study held fixed, and both land the same way. The first asks whether an agent that has noticed a record underdetermines a match can issue a **targeted request** for exactly the attribute that would settle it — the deliberate form of the discovery move above. Recognising the shortfall is easy for everyone: every agent asks rather than guesses, against a surface control pinned at the 0.50 guessing floor. Targeting it is not. The strong agent asks for the one disambiguating attribute and resolves every item (request accuracy 1.00); the mid agent usually finds it (0.89 of the time, resolving 0.94); the weak agent asks but cannot aim, requesting the wrong or a superfluous field and landing back at the guessing floor (0.50). Knowing a question is needed is not the same as knowing which question to ask. The second probe concerns the **decision to escalate**: when the cheap evidence a match needs is withheld, will the agent spend a probe — run the decisive experiment — to recover it? Denied the key, the strong agent roughly triples its probing and recovers essentially all the lost ground (recall 0.97); the weak agent barely escalates, probing about as much with the key gone as with it present, and its recall collapses by a third, to 0.46. And the escalation is reach-gated as well as capability-gated: a probe budget lifts the strong agent from 0.64 to a perfect close where a side is live to be interrogated, but buys almost nothing where the side has gone inert and there is nothing to ask. Both probes carry the caution the surprises below make general — the weak agent's apparent restraint is not judgement but an inability to aim or to escalate, conservatism rather than competence.

The lesson is as much a design one as a scientific one. How far ad hoc reconciliation carries is not fixed by the agents alone; it is set jointly by how capable they are and by how *interrogable* the thing they meet is: whether the interface on the other side lets a capable agent ask the open questions it needs to. That is a concrete thing a real deployment would have to provide, and it is exactly what the carrier and standards-body engagement discussed under scope could help pin down.

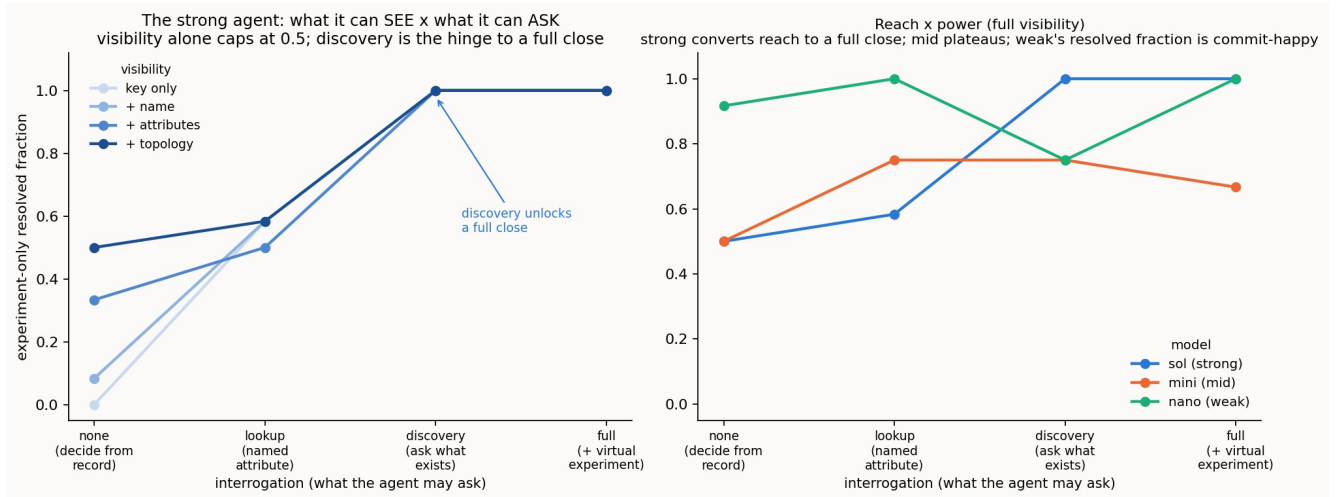


Figure 13 (reach). Left: the strong agent, with liveness held at the top. Each line is one level of visibility (how much of the record it can see); moving left to right widens what it may ask, from nothing, through asking a named attribute, to discovery (asking what exists), to a live experiment. The y-axis is the share of the hardest (experiment-only) cases it settles correctly. Seeing more (higher lines at the left) caps at one-half; discovery collapses that spread to a full close, at precision 1.00 throughout. Right: the same left-to-right axis, now one line per agent at full visibility. The strong agent reaches a perfect close; the mid agent plateaus; the weak agent's high values come with slipping precision (noted in the text), so they are not the same achievement.

13.6 The economics of the reference: when constructing it is worth the cognition

The cross-domain setting (§9) showed that where no standard exists, a capable agent can *build* the thin shared reference itself and then bind through it, lifting a stalled reconciliation to a near-complete close. Building a reference is, however, itself an act of cognition, and that cognition is not free. So a sharper question follows, and it is the one an operator would actually ask: is it worth building a reference every time, or only sometimes? To answer it we ran the same reconciliation three ways on each of the three schema settings, and (this is the new measurement) recorded not just the **close** each reached but the **cognition each spent**, counting the cost of *constructing* the reference separately from the cost of *binding* through it. The three ways are: **no reference** (the agents bind with nothing shared to lean on, the floor); **constructed** (the agents build the reference themselves, then bind, the standard-free protocol, whose cost we are weighing); and **given reference** (the agents bind through a small, already-published reference, a standard, the cheap upper bound). Two plain reminders, since they carry the result: *resolved fraction* is, of the true correspondences that exist, the share the agent finds and commits, the completeness of the close; and *reasoning tokens* are the model's hidden deliberation, a direct measure of how hard it worked.

The answer is a rule, not a blanket habit, and it has three parts. **Constructing the reference is load-bearing only where no standard exists and the agent is capable.** In the cross-domain setting (two private models, no shared standard) the two agents with no shared reference reach only a fifth of the true matches (resolved fraction 0.20, because they honestly refuse to guess a seam they cannot be sure of) and constructing the reference themselves carries them to 0.80, and to 0.90 when they also run a decisive experiment on the candidates, for a few thousand reasoning tokens of construction. There is no cheaper route, because there is no standard to bind through; the construction spend is what buys the close. **Where an effective reference already exists, constructing one is redundant.** In the configuration and observability settings, binding through the *given* reference reaches the same close for a fraction of the cognition: the strong agent binds through the given reference to a full close for a few hundred reasoning tokens, where *constructing* one spends several thousand for no better result. Building what you already have is wasted effort. **And construction is capability-gated.** A weak agent cannot build a good reference cheaply: the same non-convergence that makes weak two-agent negotiation

expensive, grinding to the round limit at tens of thousands of tokens, makes weak construction expensive too, and for a worse close. Handing construction to a weak agent is maximum spend for minimum return.

Two things this sharpens. First, it puts a number on how cheap a *given* reference is for a capable agent (tens to low hundreds of reasoning tokens to bind through) which is exactly why a pre-agreed standard is worth so much: someone paid the construction cost once, and everyone binds through it cheaply thereafter. Second, it keeps one honest caveat in view. "Redundant for this one close" is not "worthless": a constructed reference is a durable, reusable, auditable artifact, and a standard is simply a constructed reference that has been *amortised* across everyone who binds through it. So the rule has two regimes: for a one-off reconciliation between capable live agents, build the reference only where no standard exists; but at scale, or over time, building it once and keeping it pays even where any single close did not need it. That, in the end, is the argument for lightweight shared references (constructed ad hoc when none exists, reused when they do) over either a heavy universal standard agreed in advance or paying the construction cost afresh on every exchange.

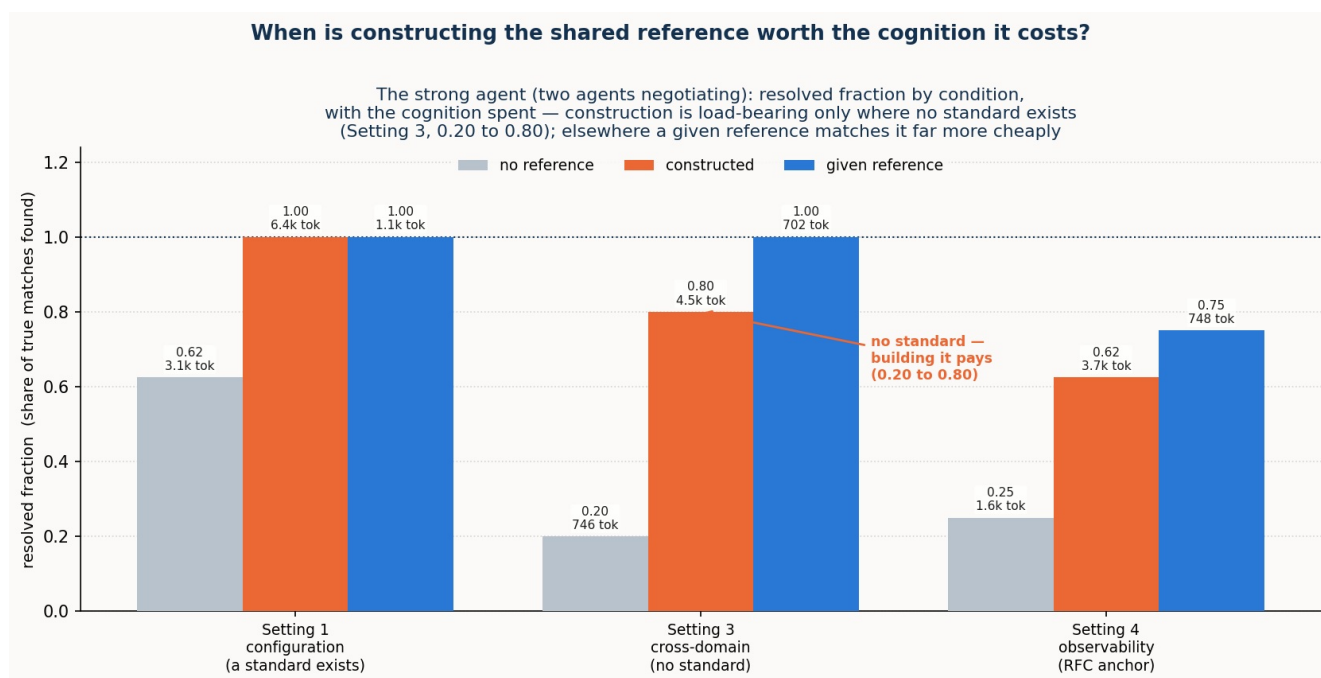


Figure 14 (the economics of constructing a reference). The strong agent's resolved fraction under three conditions, no shared reference (grey), a reference the two agents construct themselves (orange), and a given, already-published reference (blue), across the schema settings, with the reasoning each condition spent noted on each bar. In Setting 3, where no standard exists, constructing the reference lifts the two agents from 0.20 to 0.80 (0.90 with a decisive experiment on the candidates) and is the realistic path; in Settings 1 and 4 a given reference reaches the same close for a fraction of the cognition, so constructing one is redundant. Construction is capability-gated: a weak agent pays far more to build a worse reference, as the six-model ladder's non-convergence at the weak end shows.

13.7 The negotiation in the act, and the decisive experiment

The numbers above are outcomes; the process that produces them is itself a result, because it is a window into machine cognition reconciling. The box below is a verbatim excerpt, lightly trimmed, of two live agents reconciling the cross-domain seam: Agent A holding only the transport model (the `m.*` concepts), Agent B only the IP-service model (the `c.*` concepts), each seeing the other only through a surface catalogue and the answers it volunteers.

Without a shared reference — interrogate, then defer. $A \rightarrow B$: "Is *rate* the guaranteed end-to-end client payload rate, a physical bearer line rate, or an IP-service commitment?" ... (eight questions, one per concept). **$B \rightarrow A$:** *m.rate* = "the committed client payload rate guaranteed end to end by the transport circuit"; *c.rate* = "the committed information rate guaranteed to customer traffic by the IP service". **A proposes, B ratifies — one correspondence:** *c.underlay* ↔ *m.circuit*, "both denote the transport circuit itself." **Accepted.** Everything else — rate, latency, protection, the attachment point — is a transport-domain view set against a service-domain view; neither side asserts the pairing without more ground, so all are referred onward. Confirmed 1 of 5; precision 1.00, resolved fraction 0.20.

With a thin shared reference — bind through it. Same interrogation; then, each concept now carrying its binding to a shared entry, A proposes five correspondences at confidence 1.00 and B ratifies all five: *m.rate* ↔ *c.rate*, *m.latency* ↔ *c.latency*, *m.protection* ↔ *c.protection*, *m.handoff* ↔ *c.attachment*, *m.circuit* ↔ *c.underlay*. The stated rationale names exactly what the reference settled: "Although A represents a measured propagation value and B represents its SLA tier, they correspond." Confirmed 5; precision 1.00, resolved fraction 1.00.

Refused in both conditions. *m.grade* ↔ *c.grade* is never proposed: a transport quality grade (protection and restoration class) is not an IP class of service (scheduling and drop priority). The planted false cognate is declined with or without the reference.

One illustrative run of one case; the rationales are the agents' own stated reasons, presented as such.

Two things in the box are the programme's claims caught in the act. The interrogation is real, not asserted: the agents ask and answer about each concept before deciding, and because each is the authority on its own side a correspondence is confirmed only when one proposes it and the other ratifies. And the behaviour is honestly conservative: without a shared reference two careful agents confirm only the one identity they are sure of and defer the rest rather than guess across the transport/service divide, and a thin shared reference is exactly what lets them bind the pairs they otherwise defer.

Where the agents are unsure, though, they need not defer. At both-cognitive they can run a **decisive virtual experiment**: provision a candidate correspondence in virtual space, operate it, and read back whether the invariants a correct translation must preserve still hold, so the question is settled by operation rather than by argument. The box below is that experiment firing, verbatim, on the configuration case with no reference.

The decisive experiment settles what dialogue defers (configuration, no reference). Two agents reconcile TAPI ($t.*$) with TEAS ($i.*$). Where a candidate is uncertain, an agent provisions it in virtual space and reads the verdict back: **A:** provision $t.sip \leftrightarrow i.ttp?$ (does a service-interface point translate to a tunnel-termination point?) → **REFUTE [no-correspondence]** — both sides then mark those concepts no-counterpart. **A:** provision $t.cs \leftrightarrow i.tunnel?$ (does the external connectivity-service abstraction translate to the tunnel that realises it?) → **CONFIRM [identity]**. **A:** provision $t.lpq \leftrightarrow i.otnlabel?$ → **CONFIRM**; and $t.translink \leftrightarrow i.stp$ → **CONFIRM**. The agents then ratify citing the verdict verbatim — "the decisive experiment confirmed identity" — and the reconciliation closes **fully: nine confirmed, precision 1.00, resolved fraction 1.00, four experiments spent, no reference**. The correspondences dialogue would have deferred — a service abstraction against the tunnel that realises it — are settled by operation, and a look-alike is refused the same way. Given that power, two agents reach the reference's close **without a reference on three of the four schema settings**, and refute a suspected false cognate by operation. The one exception is the cross-domain seam above: there the correspondences do not exist in either agent's private model as candidates to test — they live only in a constructed shared frame — so the experiment settles the candidates the agents can pose but cannot, by itself, surface the ones a construction must first make visible. **Construction surfaces the candidates; the experiment settles them; a shared reference amortises both.** That is the completion the fully-cognitive case promises, shown rather than argued.

Measured apart, the decisive experiment and the agent's own record-reading verification prove to be genuinely different authorities, and the difference is one of *reach*. Where an independent, gold-free graph oracle can act, it is faithful — perfect where both sides are live; the agent's verifier, judging the same proposals, agrees with it about 64% of the time, and every disagreement is the agent **over-passing** a byte-clean wrong pair the experiment refutes (every split goes the oracle's way). But the oracle's authority is bought with liveness, and its reach recedes exactly as it is most needed: its coverage falls from 11 of fourteen proposals to 3 to 0 as the sides go inert, while the proposals the agent must adjudicate alone rise $3 \rightarrow 11 \rightarrow 14$. So the decisive experiment is the better authority where cognition is already strong and absent where the agent is most alone — distinct from negotiation in kind, and more sharply in reach. (An independent graph oracle is used here; a full operational emulator or digital twin, which would extend the experiment's reach into the inert placements, remains ahead.)

A matching, and deflating, result concerns the **negotiation** itself. The intent setting is framed as a two-sided negotiation, but how much of the outcome the second party's *reasoning* actually carries can be measured directly, by replacing the deterministic best-achievable oracle with an **actual** second reasoning agent and watching what moves. Almost nothing does: the live provider agent reproduces the oracle's offers (0.89 to 1.00 across the ladder) and its accept-or-refer outcomes more closely still (0.96 to 1.00), with the oracle control at 1.00. The provider behaves as a solved optimisation rather than a party whose reasoning shifts the result. On this evidence the "two-sided negotiation" language is **not earned** in the intent setting: the exchange is better read as one-sided **refinement** against a checkable oracle, with the customer's pragmatic accept-or-refer judgement (§13.4) the cognition the outcome actually turns on. It is an honest null, and it sharpens the account rather than weakening it — what looked like two agents bargaining is one agent refining against a well-specified target.

13.8 The lift, performed by an agent

Everything above runs over *lifted* models, and in the benchmark those lifts are provided as materialised artefacts (each concept's gloss and worked example, the self-explanation a cognitive side would volunteer) so the reconciliation runs over a fixed substrate. That leaves a fair question: does the result depend on the lift being authored in advance, or can an agent perform the lift itself and does reconciliation hold when it does? One half of this is already exercised elsewhere: at one-inert and both-inert a live agent reconstructs the mute side's meaning from its structure and instances, with no volunteered gloss, scored right across the spectrum. What was held fixed is the *self-lift*.

To close it, an agent is given one side's data-model surface alone (its labels, synonyms, kinds, relations, and instances, with the explanation layer, the reference binding, the other model, and the gold all withheld) and asked to produce each concept's gloss and worked example. The result is an agent-lifted model whose explanation layer was generated rather than authored, otherwise identical to the fixture. Reconciling over it, under the same conditions, and comparing to the fixture lift, measures whether the lift is agent-performable and whether the reconciliation depends on who performed it.

setting	model	no-ref: fixture	no-ref: agent-lift	ref: fixture	ref: agent-lift
configuration (flagship)	strong	0.56 (1.00)	0.56 (1.00)	1.00 (1.00)	1.00 (1.00)
configuration (hard)	strong	0.83 (1.00)	0.75 (1.00)	1.00 (1.00)	1.00 (1.00)
cross-domain	strong	0.20 (1.00)	0.00 (0.00)	1.00 (1.00)	1.00 (1.00)
observability	strong	0.25 (1.00)	0.50 (1.00)	0.75 (1.00)	0.75 (1.00)
configuration (flagship)	mid	1.00 (0.90)	0.78 (1.00)	1.00 (1.00)	0.78 (1.00)
configuration (hard)	mid	1.00 (0.92)	0.86 (1.00)	0.92 (1.00)	0.92 (0.92)
cross-domain	mid	1.00 (1.00)	0.80 (0.80)†	0.80 (1.00)	1.00 (0.83)†
observability	mid	0.50 (1.00)	0.75 (1.00)	0.75 (1.00)	0.75 (1.00)

Resolved fraction (precision in parentheses), fixture lift versus agent-produced lift, at both-cognitive, with and without the shared reference, on the strong and mid agents. Coverage 1.00 throughout: the agent glossed every concept. †the agent-lift committed one false cognate the fixture avoided (surviving false cognate = 1); false cognates were zero in every other cell.

Three things read off it. First, **the lift is always complete** (the agent produced a gloss and example for every concept on both models at both capability levels) and its **cost is capability-gated**, like everything else here: the strong agent lifts a side for a few hundred reasoning tokens, the mid agent for several thousand. Second, **reconciliation over the agent-produced lift lands in the same regime as over the fixture**, and for the capable agent binding through a reference (the mainline condition the study leans on) it is **identical across all four settings**. Without a reference the two track within the reconciliation's own single-trial variance: equal on the flagship, a little lower on the hard configuration case, higher on observability, with one soft cell at the cross-domain construct gap where the strong agent's lift led it to commit a pair the fixture deferred. Third, the divergences that do appear are **the study's own capability-gating, not a new failure**: at mid capability the agent-lifted cross-domain case slips the very false cognate the strong agent and the fixture avoid (the weak end taking a look-alike, exactly as it does everywhere) while on the configuration cases the agent's own glosses make the mid agent *more* conservative, deferring a little more and holding precision at 1.00.

A smaller point is worth keeping: the agent's glosses overlap the fixture's only weakly in wording (a lexical fidelity of 0.13 to 0.23), yet reconciliation stays in the same regime: the lift need not reproduce the authored phrasing to carry the same meaning. So the results do not rest on the lifted models being pre-authored. Who performs the lift, a materialised fixture or a live agent from the schema surface, does not change the reconciliation, most cleanly where the study leans on it; producing a good lift, like consuming one, is gated by cognitive power. What the study still holds fixed is the lift from *raw schema text* (inferring the ontology's structure, not only its explanation layer) and a cold start with no instances yet to read; both are natural experiments for real data (§16).

A follow-up widens this from the two measured agents to the full capability ladder. Lifting alone, with no reconciliation, across all eleven schema cases and three models (a strong, a mid, and a deliberately weak one), every lift came back complete and structurally identical to its fixture, and the false-cognate traps were caught at every rung: even the weak model read the tunnel-termination-point as a trail head rather than a link end, and, in the relabelled-identity control where the recognisable names are stripped away,

still told the concept that *originates* a conduit from the one that *terminates* a span, reasoning from the relations alone. What capability moved was not correctness but verbosity: the weak model wrote about fifteen words of gloss per concept, the strong twenty-two, the mid twenty-six, and, against first intuition, the tersest model tracked the (terse) fixture wording most closely. This is a lift-only spot-check, one trial per case, but across eleven cases the reading is steady: the structural layers of the lift are recovered well at every capability, and the one thing recovered at no capability is the pragmatic layer. That is not a shortcoming of the lift, and it is the more important half of the story, so it earns its own reading.

The box below is the lift caught in the act, on the relabelled-identity control, where the recognisable names are replaced by invented ones so that nothing but the structure is left to reason from. Asked to record, per concept, the surface signals it used and the look-alike it ruled out, the strong agent reports (its own stated reasoning, with concept ids rendered as their names):

The lift in the act (relabelled control, strong agent).

flux-span , kind link; between two vertices, terminated-by a cap-point; instance *fluxspan-R1-R2* . "A topology link; ruled out the flux-conduit, which is a service over the wider fabric rather than a single inter-vertex link." → **a direct link between two vertices.**

cap-point , kind termination; on-vertex , terminates the flux-span; instance *cp-R1-R2* . "A span termination; ruled out the conduit-cap-point, which originates a conduit rather than terminating a span." → **the point where a span ends at a vertex.**

conduit-cap-point , kind termination; on-vertex , originates the flux-conduit; instance *ccp-A1* (conduit head) . "A conduit-origin termination; ruled out the cap-point, which terminates spans, and the conduit-circuit-endpoint, which is an endpoint belonging to a conduit." → **the head from which a conduit originates.**

flux-conduit , kind service; from the conduit-cap-point, over the fabric; instance *conduit-ala3-grd2* . "A network service; ruled out the flux-span, a single link rather than an end-to-end service." → **a service that originates at the conduit head and is carried across the fabric.**

The two look-alike terminations are told apart by a single relation: the cap-point terminates a span, the conduit-cap-point originates a conduit. The reading of the conduit then leans on the head that originates it, so a lift is relational, not concept-by-concept in isolation. Nothing here is recognition: the names are invented, and the readings are built from kinds, relations, and instances alone. The evidence is the agent's own stated reasoning, shown as such.

Two things in the box are worth drawing out. The reading is genuinely structural, not a memory of a known standard, which is what the relabelling was built to test: with *span* and *conduit* and *cap-point* carrying no public meaning, the only thing separating the two terminations is that one *terminates* and the other *originates* , and that is exactly the distinction the agent names. And the lift is relational: to read the conduit the agent leans on the head that originates it, and to read the span it leans on the termination that anchors it, so the flow of a lift runs along the model's own relations rather than concept by concept. This is the schematic layer being recovered by reasoning, and it is the counterpart, for the lift, of the negotiation and the decisive experiment shown earlier for reconciliation.

13.9 Pragmatics across the lifecycle: what the lift holds, and what only a live operation can

The result just seen, that an agent lifting from the bare schema recovers the structure but never the pragmatic layer, is easy to read as a limitation and is not one. It points instead to a distinction inside the semantic model that is worth drawing precisely, because it is the distinction a modelling tradition centred on the schema most easily misses: pragmatics are not a harder part of the ontology, they are a different kind of thing.

The cleanest way to tell a lifted model's parts apart is not by what each contains but by how a consumer

obtains it, and doing so separates the ontology's schematic and concrete layers (§2) and sets the pragmatics clearly apart from both. The **schematic layer** is the concepts, their kinds, and their relations, the TBox: a standing structure that changes only when the model is re-lifted. The **concrete layer** is the instances that populate it now, the ABox: dynamic, but obtained by **observation**, a determinate re-reading of the live source that even a non-cognitive consumer can perform. The **pragmatics** are neither a schema axiom nor a recorded fact. They are the interpretation over both: whether a state matters, whether a degraded offer is acceptable, whose authority governs a contested field, what a correct translation must preserve. They are obtained not by lookup but by **judgement**, inferred from the schema and the instances and the current context, defeasible, and context-relative. So the pragmatic layer is not the dynamic cousin of the ABox. An alarm's current state is *read*; whether it warrants action here and now is *judged*. The ABox is observed; pragmatics are resolved.

That distinction fixes exactly what a lift can hold. It holds the schematic layer in full, because that is structure, and structure is what the agent-lift recovers from the surface at every capability, even with the names stripped. It holds the concrete layer as a **snapshot**, which goes stale, but staleness is cheap: anyone can refresh it by re-observing the source. The pragmatic layer it can only **describe**. An owner that knows its context can write down the rule and its dependency, "the concern score is recomputed from the current context, and is lower during a maintenance window," but it cannot freeze the resolved value, because that value is a judgement against a context not fixed at lift time. A stranger lifting from the bare surface cannot even describe it, because the rule is not in the structure, which is precisely what the eleven cases show: the pragmatic nuance absent at every rung, and rightly so, since to invent it would be to fabricate it. The lift, then, holds the schematic layer fully, the concrete layer as a refreshable snapshot, and the pragmatic layer only as a standing description of a situated thing.

This is not an artefact of our JSON encoding; a standard ontology has the same limit. The content of a lifted concept maps cleanly to OWL:

lifted-model field	OWL / RDF
concept	<code>owl:Class</code>
label	<code>rdfs:label</code>
synonyms	<code>skos:altLabel</code>
kind	a class annotation
gloss	<code>skos:definition</code>
example	<code>skos:example</code>
ref	<code>skos:closeMatch</code> to a reference entry
relations	typed object properties
instances	<code>owl:NamedIndividual</code> , typed to the concept

Every structural field has a home. But the pragmatic content lands only as inert annotation text. The observability concern score becomes a class carrying a definition string:

```
:concern-score a owl:Class ;
  rdfs:label "concern-score" ;
  skos:definition "a dynamic concern score, recomputed from the current
    context, lower during a maintenance window" ;
  rdfs:subClassOf [ a owl:Restriction ; owl:onProperty :annotates ;
    owl:someValuesFrom :anomaly ] .
:concern-78 a :concern-score .
```

A description-logic reasoner can act on the class and on the individual `concern-78`, but not on

"recomputed from the current context": there is no TBox axiom and no ABox assertion for a defeasible, context-relative judgement. OWL holds the schema, the data, and a description of the pragmatics; the pragmatic judgement itself has no native representation in it. The limit belongs to the pragmatic layer, not to the format: any static artefact, our JSON, an OWL graph, or a YANG module, can carry a description of a pragmatic rule but not its resolution.

If no lift can hold the resolved pragmatic layer, then something must supply it, and that something is the **live operation** over the model. Reconciliation is the operation this study measures, but the mechanisms are general, and each one appears in more than one setting. There are four. An operation can **pull in** a pragmatic fact the lift never carried, by interrogating a live peer: the open question "what facts do you have?" is the study's decisive lever, and it is asked in reconciliation, in intent (of the operator's catalogue), and in observability (of the resource). It can **refresh** a dynamic pragmatic value, by re-observing the current context and re-resolving against it: the concern is low because the maintenance window is open *now*, a reading the observability significance and the intent lifecycle both make. It can **create certainty** where the lift and the dialogue leave a candidate uncertain, by the decisive virtual experiment: provision the candidate in virtual space, operate it, and read back whether the invariants a correct translation must preserve still hold, which is the same mechanism as intent's satisfaction check and a correlation test in observability. And it can **resolve** the pragmatic verdict itself, the significance, the acceptability, the authority call, by judgement against the current context, which is the capability-gated frontier of Thesis 5 and the decisive operation of settings 4, 2, and 3 in turn. These are operations over any lifted model, not tricks of reconciliation; reconciliation is only where we have instrumented them most fully.

Seen this way, the static artefacts of the study are not rivals to the live operation but **frozen outputs** of it. A published reference or lexicon is a frozen *pull-in*: a description someone interrogated out of a system earlier. A snapshot, the ABox the lift holds, is a frozen *refresh*: an observation taken at lift time. And a pre-placed policy, the movable policy of the intent setting, is a frozen *resolve*: a pragmatic verdict judged once and carried forward. Each stands in for its live mechanism, and each holds only until the context it assumed drifts, which is exactly the point at which the live mechanism would have had to run again. You can freeze a past judgement, but not the judgement for a context not yet met, and that is the irreducible reason a static model can describe pragmatics but never carry them resolved for an unforeseen case. This also fixes the behaviour across the cognition spectrum, because the mechanisms are gated by two things at once. The first is **liveness**: pulling in needs a live peer to interrogate, refreshing a live source to re-observe, creating certainty a live substrate to experiment on, so as sides go inert these fall away in order, until at both-inert only the frozen description survives and every pragmatic verdict must be referred onward. The second is **capability**, which bites hardest on resolution: even with a live side present, pulling in and refreshing are close to mechanical, while judging significance, acceptability, or authority is the capability-gated frontier. The two gates are why the observability verdict is settled only by a capable agent with a live side to read, while correlation, a structural test, holds across the whole ladder. The framework does not merely extend to the operations beyond reconciliation; it predicts their signatures.

Put individually under test, the three operations bear the prediction out, each with its own regime.

Pragmatic resolution — the significance verdict — delivers a large gain for capable agents and almost none for the weak one: +0.33 and +0.58 verdict accuracy for the strong and mid agents, +0.03 for the weak. The operation is real and the context genuinely changes the right action, but only an agent strong enough to carry the pragmatics collects the benefit. **Attribute pinning** — fixing a shared field's operative meaning, a committed rate against a line rate — runs the opposite way, as a weak-model *compensator*: it buys the strong agent essentially nothing (it already drew the distinction) and the weak agent about +0.19, standing in for a discrimination it could not make itself. And **correlation** — grouping reconciled parts into wholes, several symptoms into one incident — is the structural input the framework predicts is robust: unconditionally needed (with it off, the two weaker agents resolve none of the multi-symptom structure) and, once on, holding across the whole ladder. Two further results show correlation is a genuine, portable operation rather than an observability artefact: it returns the same partition-exact

result in the configuration setting, so it *generalises* beyond the setting it was found in; and it survives a **self-derived** dependency map — every agent derives the map perfectly from raw inventory, a bundle indirection and a name lure included, so correlation on the derived map matches the given-map baseline. The structural pragmatic does not depend on a hand-fed input any more than the descriptive reference did.

The whole picture, then, is this. The lift is powerful and, as the eleven cases show, robustly performable by an agent for the layers that are structural, at every capability. But the pragmatic layer is not something a lift, however careful, or an ontology, however standard, ever holds resolved. It is pulled in, refreshed, made certain, and resolved by a live cognitive operation reading the current context. That is why performing the reconciliation live is not an overhead to be optimised away but the very mechanism by which meaning that no artefact can hold is supplied on demand; it is why the pragmatic layer is the frontier the descriptor methods never reach; and it is the half of the semantic model that a schema-centred tradition most easily overlooks, because it does not live in the schema at all.

13.10 Producing the inputs: alignment, reuse, and derivation

A reconciliation needs inputs — two lifted models, and often a shared reference or a settled authority to bind through — and a fair challenge is whether the study has shown only reconciliation *given* those inputs while the genuinely hard part is producing them. Put under test, the inputs prove more often *producible* than assumed, under the same capability caveat that governs everything else.

Aligning two **independently authored lexicons** — no reference id shared across sides, the harder half of the schema bind — is tractable for a capable agent: it holds precision 1.00 by aligning the two vocabularies on their meaning, where a shared reference would have supplied the alignment for free (§13.3). The hard half is doable; it is doable by cognition, and it is capability-gated.

Reuse turns the scaling argument from a structural count into a measured economy. Onboarding a genuinely new system against an established shared reference is linear where pairwise integration is quadratic, and the effort bears it out: reuse is 2.0 to 2.7× cheaper in reasoning tokens *and* holds recall where the pairwise route lets it slip — cheaper and more accurate at once. Thesis 6's scaling claim is now exercised on real effort, not only counted on the interoperability graph.

And the two structured inputs a reconciliation leans on turn out to be **derivable** rather than necessarily hand-authored. The observability **dependency map** the correlation operation runs on can be derived by an agent from raw inventory through a multi-hop join — resolving a bundle indirection and refusing a name lure — and correlation on the derived map matches the given-map baseline (§13.9). Cross-domain **authority**, whose realm owns a contested field, can be derived from an external artefact such as an interconnect agreement: structure-only attribution is sharply capability-gated (a correct attribution 1.00 of the time for the strong agent, 0.47 and 0.40 for the mid and weak, the weak agent taking a transport lure), but the artefact lifts every agent to a full, correct attribution while the agent still defers on the field the artefact leaves genuinely co-owned. And when two authority sources *contradict* each other, every agent along the ladder **surfaces** the conflict rather than silently choosing a side — the failure one might fear does not occur when the disagreement is visible.

The line these results draw is sharper than "reconciliation works given its inputs": producing the inputs is itself often achievable, an external artefact or reference is the reliable route to it, and the residual risk is the capability-gated confident error the rest of this section has traced — which is why an abstain path and a floor on reference quality matter as much as the reconciliation they protect.

14. The maps

The four readings above are the evidence; the two tables below are the index to it. The first locates every finding on the **cognition spectrum** (the master variable) and the second on the **operation** it acts on, so

that "cognition matters here" always resolves to a named row.

Table 1: the cognition spectrum, and what happens at each placement. The rows are the master variable; the columns say what the reconciliation can do, what it leaves in the residual, and what a reference can buy, consolidated across all four settings.

placement	what the reconciliation can do	the residual	what a reference buys
both-cognitive	completes autonomously : bind (via a shared/constructed reference or a decisive virtual experiment), negotiate, verdict	none once a reference or experiment closes it; dialogue alone defers the hardest	speeds the close and can substitute for reasoning; but the negotiation itself already holds precision and catches the traps
one-inert	live side reconstructs the mute side; probes gone, so some resolutions turn structural	grows: the shortfall appears	supplies missing information ; rescues the mid agent (ontology, satisfaction)
both-inert	third party can only propose for adjudication; no interrogation, no experiment	largest; every judgement referred onward	supplies information, not authority ; cannot move a judgement
pre-placed policy (intent)	closes the decisions the policy was authorised for: pushes the hand-off boundary outward	only the un-authorised judgements	policy carries the authority ; a reference still cannot

Table 2: the operations of a reconciliation, and where each was put under test. The rows are the process stages of §3; a cell says how the setting exercised that operation, or "—" where it did not. This is the map for locating any finding on the process: the *here* of "cognition matters here" is a row.

operation (the <i>where</i>)	1 · Configuration	2 · Intent	3 · Cross-domain	4 · Observability
Lift	founds it; the lift is the lever (controls: surface ~two-thirds, cognition closes the rest)	reused	reused; both sides private	alarm lifted and decomposed
Reference construction	given (lexical)	given (unit / invariant)	constructed end to end: 0.20 → 0.80 (0.90 with a decisive experiment)	given (RFC 9940-anchored)
Schema binding (lexical, ontological)	equivalence; reference substitutes	shared ground for satisfaction	the measured stage: omission vs commission	ontological cognate; 3-rung gradient
Attribute pinning	—	bound vs measured metric	committed vs line rate	severity / scores
Instance co-reference	budget-limited → structural	measured (endpoints)	bracketed (reproduces s.1)	measured; reproduces s.1
Verification	ratification + decisive experiment; catches cognates, holds precision	by satisfaction	downstream of the bind	the verdict carries it
Pragmatic resolution	left untouched (deferred)	movable policy ; authority ≠ information	authority attribution; reference pins meaning, not authority	verdict carries operative meaning; capability-gated
Composition / correlation	—	—	—	robust with the dependency map (all models)
Lifecycle recurrence	—	four-hop loop (self-heal, refer, restore)	—	—
Scaling	linear vs quadratic in the number of systems	—	—	—

Two rows carry more than their per-setting cells show. The **Lift** is not only the lever but is itself agent-performable, and robustly so: an agent lifts each side from its schema surface alone, recovering the ontology at every capability and even with the names stripped (§13.8). And **Pragmatic resolution** is the one operation obtained by judgement rather than lookup: a lift can describe a pragmatic rule but never resolve it, so the resolution is pulled in, refreshed, and settled by the live operation and is never held by any static model (§13.9).

15. The surprises

Five results ran against the naive expectation, and they are worth stating as findings rather than smoothing away.

A strong agent scoring *lower* is often a strong agent behaving *better*. Blind at both-inert, the strong agent refuses to affirm what it cannot verify and scores below the mid agent (setting 2's

satisfaction 0.29 vs 0.71; setting 3's under-commitment). This is not weakness; it is the honest deferral of Thesis 4, and it is exactly the behaviour a published reference or a live probe converts into a confident, correct close. A raw accuracy number, read without the precision beside it, misjudges it.

A bare shared pointer is worse than no reference at all. One might expect any shared anchor to help; setting 3's ablation shows an opaque identifier with no description dropping precision below the no-reference floor (0.50), because the agent binds by the token and binds wrongly. The reference works through shared *description*, never through the pointer, and the corollary (setting 1) is that a shallow *class* tag, the thinnest description, can mislead the weak agent it was meant to help.

The richer content that raises a weak agent's reach also raises its exposure. The lifted content and instances that let an agent find more correspondences also hand a weak agent more surface to misfire on: its instances turn toxic, over-read as evidence of identity (setting 1's baseline). The lift is unambiguously the lever and unambiguously safe for the strong agent; for the weak agent it is double-edged.

Two operations in the same setting place opposite demands on the agent. In observability, the verdict (a judgement over concern, confidence, and context) is sharply capability-gated, while the correlation (an application of a dependency graph) is robust across the whole ladder. Both resolve significance from the same pragmatic information and both are decisive, but one demands judgement and one is a structural input, and that distinction, not the label "pragmatic," predicts whether a weak agent can do it.

A "capability-independent" law turned out to be one model family's habit. The verifier's apparent ceiling — a pass rate near 0.71 that did not move with capability — looked like a structural property of record-only verification. Replicating across three model families showed it was not: each family verifies with its own disposition (one permissive, the others less so), and capability does not explain the spread (§16). The cross-framing hardness a separate grader finds, by contrast, is family-independent. The lesson is a general one: a single family can make a disposition look like a law, and only replication across families tells the two apart.

16. Scope, threats, and what remains

The claims here are **existential and mechanistic** (*this is how ad hoc reconciliation works, and here it is working*), not population estimates. Each setting is built around seeded cases, designed to exercise each mechanism and prove each trap rather than sampled from a distribution, with a small number of trials; the reported patterns are the ones stable across the model ladder and the treatment toggles, and the numbers are indicative rather than tight. To guard against any one case being, unknowingly, chosen to work, each setting is exercised on **three independently-built cases**, not one; the breadth reading at the end of this section reports that the findings hold across them. The core ladder is three points spanning a capability range; a six-model sweep at both-cognitive shows how two-agent negotiation shapes the gradient: bilateral ratification holding precision and suppressing false cognates across the ladder, resolution lower and non-monotonic (§13.2). Golds are derived from the models and validated (internally consistent, and for the flagship case independently corroborated by a separate grader that rejects the same trap and cross-framing pairs across two model families — see the cross-family reading later in this section), which removes drift but leaves the modelling choices (including setting 4's verdict thresholds, stated openly) as authored rather than found. Public standards may have been seen in training, which could flatter the no-reference conditions; a **relabelled-identity control** isolates this directly: strip the TAPI/TEAS identity from the flagship case and reasoning alone reaches a resolved fraction of 0.67 where the recognisable case reaches 1.00, while a shared reference restores 1.00 either way, so recall of a known standard accounts for part of the recognisable close but the reasoning is real and a reference substitutes for the recall. The effect relied on throughout is the *difference* a treatment makes under identical inputs.

A single model family is itself a threat, and replicating across families both confirmed the findings and corrected one. Every result above uses one lineage, the gpt-5.x ladder, so the sharpest external question is whether a finding is a property of ad hoc reconciliation or of that family. Re-running the sharp experiments against two other families and tiers — an open-weights Qwen-2.5-7B served locally, and DeepSeek at the strong tier, reached through the same OpenAI-compatible client with a JSON-mode compatibility shim and at no cost beyond a few dollars of API — answered both ways. Most findings transferred: contradictory authority sources are surfaced by every model of every family, and the external-artefact remedy rescues every family and tier (Qwen 0.47 → 0.93, DeepSeek 0.40 → 1.00, matching the gpt-5.x mid and weak rungs lifting to a full close). But one apparent law did not hold, and its failure is the most valuable thing the replication produced. The record-reading verifier's pass rate near 0.71 had looked **capability-independent** — identical across the whole gpt-5.x ladder, seemingly a ceiling set by what the records make visible. Across families it is not a ceiling at all: the three families verify with three dispositions — gpt-5.x permissive (over-passing byte-clean wrong pairs four times of four), DeepSeek intermediate (2.7), Qwen conservative (zero) — giving accuracies 0.71, 0.76, 0.84, and capability does not order them, since a strong different-family model and a weak one *both* over-pass less than the entire gpt-5.x ladder. What read as a structural law was one family's disposition; only replication tells the two apart. Separately, an **independent grader** drawn from a second family, asked to re-judge the derived gold, rejects the same trap and cross-framing pairs the first does — so the gold's negatives are trustworthy across families and its harder merges are the same identifiable pairs under any capable judge — while a grader too weak for the task cannot perform the role at all, the capability-gating story reappearing in the grader's own seat. The honest bound is that this establishes cross-family *transfer*, not a full cross-family ladder: the breadth spans open mid-tier models plus one strong non-OpenAI model, not strong-tier breadth across many lineages.

Two operations bear noting on how they are measured. Setting 3's schema-binding headline brackets the reference-construction step, isolating the worth of that step; the full **construct-then-bind** protocol (the agents building the reference themselves and then closing, with none pre-given) is also run, and it confirms the thesis in the hardest setting: the strong agent lifts from a no-reference resolved fraction of 0.20 to a constructed-reference **0.80** (0.90 with a decisive virtual experiment on the candidates), approaching the reference-given 1.00, at perfect precision and with no false cognate, so constructing the shared ground works and building it is the work (§9). And **instance-level co-reference** is measured in settings 2 and 4 as well as setting 1, reproducing the same budget-limited-then-structural pattern, with a capability gradient in which capable agents resolve fully where a live side can be interrogated while the weakest agent only partially resolves. And the **lift** itself is measured as an agent act, not only assumed: an agent producing each side's explanation from its schema surface reconciles as the authored lift does (§13.8), identically for the capable agent through a reference, so the results do not rest on the lifted models being authored in advance; a spot-check across eleven cases and the full capability ladder finds the lift's structural correctness holding at every rung, with capability moving verbosity rather than accuracy. What stays fixed on this axis, and is natural real-data work, is the lift from *raw schema text* (inferring the ontology's structure, not only its explanation layer) and a **cold start** with no instances yet populated to read.

The single-case worry, answered as far as in-house work can. The sharpest threat to a mechanistic claim built on one case per setting is that the case was, unknowingly, chosen to work. To test that, two further cases were built for every setting (deliberately different in domain, vocabulary and traps) and the same agents were run on them under the same harness, so that three independent cases now stand behind each setting. The findings reappear on the new cases (Figure 15). In **configuration**, the strong agent again reconciles two new pairs of standard models on its own, and the thin shared reference again mainly serves to prevent the weaker agents' errors. In the **cross-domain** setting, the mirror returns on two new pairs of private, no-standard models: without a shared reference the strong agent under-commits (perfect precision on what it binds, but low resolved fraction because it refuses to guess the seam), and the constructed reference completes the close. In **intent**, working out which offers meet a customer's wish and deciding accept-or-refer under a policy again complete for capable agents, while the

multi-hop service lifecycle again grades with capability. In **observability**, the deep alarm-versus-anomaly look-alike is again reliably avoided across two new fault domains. This moves each result from "here it is on one case" to "here it is again on cases built to be different". It does not, and cannot, stand in for real network data: the same hand built the new cases too, so they test robustness to *variation*, not *realism*.

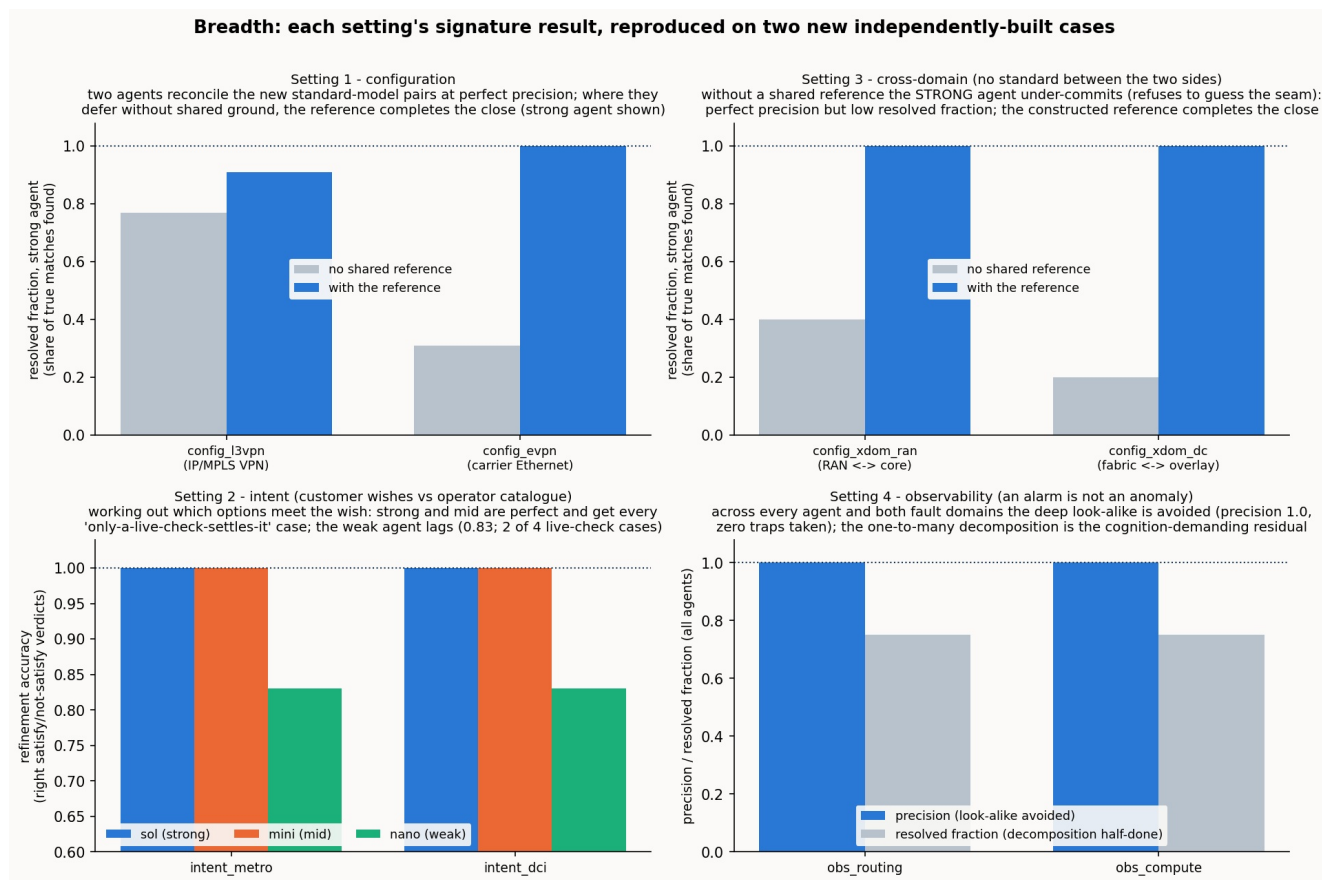


Figure 15 (breadth). One panel per setting. Each shows the setting's signature result on the two new cases built for it, using the real agents scored against the validated answer key. Setting 1: the weaker agent's precision (share of committed matches that are correct) recovers to a clean close once the shared reference is added. Setting 3: the strong agent's resolved fraction (share of true matches found) is low without a reference (it is refusing to guess) and completes with one. Setting 2: the agents' accuracy at working out which offers satisfy the wish, high for the strong and mid agents and lower for the weak one. Setting 4: precision stays at 1.0 (the look-alike is never taken) while resolved fraction sits at 0.75 (the one-to-many decomposition is the residual). The point of the figure is not any single bar but that all four signatures recur on cases built to differ.

One further case pushes past controlled variation toward the kind of mess real data brings. A deliberately **structurally messier vendor model** was built — the same service reconciled against a vendor dialect that buries the correspondences under proprietary containers and augmentations and plants a leafref false cognate — and run under the same harness. Surface matching collapses on it, losing two-thirds of the correspondences (recall 0.33 against 0.5 on the tidy cases); cognition absorbs the mess, the agents recovering to 0.89-1.00 at precision 1.00, resolving the configuration-versus-state distinction the vendor structure obscures, and never taking the leafref cognate. It is one reduced case, not a production module, but it is the first evidence the approach survives structural messiness and not only vocabulary variation — the direction real vendor data would push hardest.

What remains is therefore genuinely external: larger and more varied cases drawn from real networks, and the real-data grounding that only carrier and standards-body collaboration can supply.

17. In one paragraph

Two systems that can reason do not need a standard agreed in advance to work together; they can lift their data into portable, self-describing semantic models and reconcile those models ad hoc, for the occasion. The lift that produces those models is itself an act of cognition an agent can perform, recovering the ontology from a bare schema surface at every capability, even with the names stripped. Across four settings (configuration, intent, cross-domain, and observability) that is what happens: at the fully-cognitive end the reconciliation completes autonomously in every case, negotiations and significance verdicts included, with no standard and no human. Cognition is what completes it; descriptor methods carry it most of the way, then stop, and the placement of cognition governs how much of the remainder is closed by machine and how much is honestly referred onward. A thin published reference earns its place inside this frame (partly substituting for a capable agent's reasoning, supplying a live agent the facts it can no longer get from an inert side, and, across many systems, making the work grow linearly instead of quadratically) but its reach ends at information: it never provides the authority to decide (whose value governs, whether a trade-off is acceptable), and it must carry meaning rather than a bare pointer to help at all. And the pragmatic layer (what a reconciled thing is for, whether it matters, who decides) is the frontier the descriptor methods never reach, and the one layer no static model holds: obtained by judgement rather than lookup, it is pulled in, refreshed, and resolved by a live cognitive operation against the current context, carried by that cognition or by cognition pre-placed as a policy, and realised only by an agent capable enough to carry it. That pragmatic layer, and the question of how capable an agent must be to work in it, are where the next work lies.